DEEP RESEARCH DOSSIER (OLD) • DETERMINATIVE V5.3

Sample: Melinda Ehrle (Melinda_Ehrle) | Reference: GRCh38.p14 | Modality: 33.4x BAM mean depth (Median 30.0x) WGS (panSN GBZ aligned)

This clinical genomics evidence synthesis evaluates a broad exploratory screening corpus of 1,653 prioritized candidate variants across 1,129 genes derived from a **33.4x BAM mean depth (Median 30.0x)** whole-genome sequencing (WGS) pipeline for Melinda Ehrle. To prevent cognitive bias and diagnostic over-interpretation, this report enforces rigorous domain segregation across **Monogenic Pathogenic Alleles**, **Actionable Pharmacogenomics**, **Secondary Organ-System Modifiers**, and **Endogenous Protective Factors**.

Gene	Variant	Zygosity	ClinVar	CADD (Phred)	REVEL	AlphaGenome AVI (Phred)	Associated Condition
POLG	p.Gly737Arg · c.2209G>C · rs121918054	Heterozygous	Pathogenic/Likely pathogenic	26.3	0.936	**30.3 (AlphaMissense)**	POLG · Mitochondrial DNA Maintenance Medication Safety [VCV13513]
TAT	p.Arg57Ter · c.169C>T · rs118203914	Heterozygous	Pathogenic	36.0	—	**38.8 (Protein Termination)**	TAT · Hepatic Tyrosine Metabolism (Tyrosinemia type II) [VCV402]
ATM	c.1236-2del · rs1565381646	Heterozygous	Pathogenic/Likely pathogenic	—	—	**—**	ATM · DNA Repair & Hereditary Cancer Screening [VCV574589]
VDR	c.-1172A>G · rs4516035	Heterozygous	Likely pathogenic	5.4	—	**2.1 (DNASE-seq)**	VDR · Vitamin D Signaling & Bone Health [VCV333665(

Gene	Diplotype	Phenotype	High-Risk Medications	Clinical Prescribing Guidance
POLG	`p.Gly737Arg · c.2209G>C · rs121918054`	Mitochondrial Carrier	**Sodium Valproate (Depakote)**	**ABSOLUTE CONTRAINDICATION:** High risk of fatal valproate-induced hepatic necrosis. Avoid valproate in all forms.
ABCG2	`*ref/*rs2231142 (Q141K (rs2231142) G>T)`	Decreased Function	**Allopurinol, Clozapine, Imatinib, Methotrexate**	Allele G is associated with decreased dose of allopurinol or febuxostat in people with Gout as compared to allele A.
CYP2C19	`*2/*2 (splice defect (rs12769205) A>G, splice defect (rs4244285) G>A)`	Poor Metabolizer	**Clopidogrel (Plavix)**	Significantly reduced active metabolite formation and high risk of cardiovascular ischemic events/stent thrombosis. Avoid clopidogrel; prescribe alternative antiplatelet agent (Prasugrel or Ticagrelor).
CYP3A43	`*2/*3 (Y25fs (rs61469810) delA, P340A (rs680055) C>G)`	Unknown	**CPIC Substrates**	Titrate dosing per CPIC.
CYP3A5	`*3/*3 (splice defect (rs776746) T>C)`	Poor Metabolizer	**Tacrolimus (Prograf)**	CYP3A5 *3/*3 non-expressor (Poor Metabolizer). Requires standard lower initial dosing (0.10-0.15 mg/kg/day) compared to CYP3A5 expressors to achieve target trough concentrations. Monitor blood levels (TDM).
CYP4F2	`*2/*3 (V433M (rs2108622) C>T, W12G (rs3093105) A>C)`	Altered Enzymatic / Pathway Activity	**CPIC Substrates**	Titrate dosing per CPIC.
NAT2	`*4/*7 (R268K (rs1208) G>A)`	Intermediate Metabolizer	**CPIC Substrates**	Titrate dosing per CPIC.
SLCO1B1	`*5/*37 (N130D (rs2306283) A>G, V174A (rs4149056) T>C)`	Decreased Function	**CPIC Substrates**	Titrate dosing per CPIC.
VKORC1	`*H6/*H6 (regulatory/intronic (rs7294) C>T)`	Altered Enzymatic / Pathway Activity	**Warfarin (Coumadin)**	Carries VKORC1 -1639G>A sensitivity allele (rs9923231 variant). High pharmacological sensitivity to vitamin K antagonists requiring lower initial and maintenance doses to prevent bleeding complications.

ANALYTICAL CALLSET CONFIDENCE

98%

High-depth 33.4x BAM mean depth (Median 30.0x) WGS callset aligned against panSN GBZ pan-genome graph; >99.8% callable SNV precision.

PRIMARY PATHOGENIC CARRIER STATUS (POLG, TAT, ATM, VDR)

95%

Orthogonal concordance between ClinVar Pathogenic curation records, CADD Phred scores, and AlphaGenome AVI scores.

CLINICAL DOMAIN GROUPINGS • LANDSCAPE LAYOUT

Organized into distinct clinical categories. Formatted with page breaks and landscape orientation for clean multi-page review.

1. Pathogenic & Monogenic Carriers

5 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessio
ATM	c.1236-2del	chr11:108250699	rs1565381646	Heterozygous	**Tier1**	Pathogenic/Likely pathogenic	—	—	—	VCV5745
CDKN2B	c.*2619C>T	chr9:22003368	rs1063192	Heterozygous	Tier3	Likely pathogenic|protective	0.2	—	1.3	VCV8126
POLG	p.Gly737Arg	chr15:89323460	rs121918054	Heterozygous	**Tier1**	Pathogenic/Likely pathogenic	26.3	0.936	30.3	VCV13513
TAT	p.Arg57Ter	chr16:71576247	rs118203914	Heterozygous	**Tier1**	Pathogenic	36.0	—	38.8	VCV402 /
VDR	c.-1172A>G	chr12:47906043	rs4516035	Heterozygous	Tier3	Likely pathogenic	5.4	—	2.1	VCV3336

2. Actionable Pharmacogenomics & Drug Response

66 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessio
ABCB1	c.2065-76T>A	chr7:87544351	rs1922242	Heterozygous	Tier3	drug response	0.7	—	0.8	VCV8295
ABCB1	c.2064+73A>G	chr7:87544750	rs2235046	Heterozygous	Tier3	drug response	0.5	—	5.8	VCV8287
ABCB1	c.1725+38G>A	chr7:87549310	rs2235013	Heterozygous	Tier3	drug response	6.4	—	1.4	VCV8286
ABCB1	c.1554+81C>T	chr7:87549770	rs2235035	Heterozygous	Tier3	drug response	2.0	—	1.1	VCV8294
ABCB1	c.1554+24T>C	chr7:87549827	rs2235033	Heterozygous	Tier3	drug response	1.8	—	0.5	VCV8294
ABCB1	c.1000-44G>A	chr7:87550882	rs10276036	Heterozygous	Tier3	drug response	0.2	—	0.6	VCV8294
ABCB1	c.828-106A>G	chr7:87554038	rs1922240	Heterozygous	Tier3	drug response	5.1	—	2.7	VCV8286
ABCB1	c.827+649A>C	chr7:87560614	rs868755	Heterozygous	Tier3	drug response	1.0	—	0.1	VCV8294
ABCB1	c.703-966G>C	chr7:87562353	rs13237132	Heterozygous	Tier3	drug response	0.8	—	8.7	VCV8286
ABCB1	c.702+280G>A	chr7:87565790	rs1202170	Heterozygous	Tier3	drug response	0.2	—	0.1	VCV8294
ABCB1	c.530+139C>T	chr7:87566646	rs1202168	Heterozygous	Tier3	drug response	3.1	—	6.4	VCV8286
ABCB1	c.338+306C>T	chr7:87569866	rs1016793	Heterozygous	Tier3	drug response	0.2	—	1.5	VCV8286
ABCB1	c.338+224G>T	chr7:87569948	rs2235021	Heterozygous	Tier3	drug response	0.7	—	3.1	VCV8285
ABCB1	c.338+223A>T	chr7:87569949	rs2235020	Heterozygous	Tier3	drug response	1.2	—	1.8	VCV8285
ABCB1	c.286+553T>G	chr7:87584959	rs7809208	Heterozygous	Tier3	drug response	1.6	—	1.2	VCV8285
ABCB1	c.118-118T>G	chr7:87585798	rs1211152	Homozygous	Tier3	drug response	0.3	—	0.7	VCV8285
ABCB1	c.-1A>G	chr7:87600185	rs2214102	Homozygous	Tier3	drug response	7.0	—	5.1	VCV8293
ACP1	c.231+53C>T	chr2:272203	rs11553746	Heterozygous	Tier3	drug response	14.7	0.143	24.3	VCV3900
ADRB2	p.Gly16Arg	chr5:148826877	rs1042713	Heterozygous	Tier3	drug response	17.9	0.079	19.5	VCV2259
APOE	c.-286T>G	chr19:44905579	rs405509	Heterozygous	Tier3	drug response	10.2	—	5.2	VCV61995

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessio
APOE	c.43+520G>A	chr19:44907187	rs769450	Heterozygous	Tier3	drug response	4.0	—	6.1	VCV61995
APOE	c.*1944T>C	chr19:44911194	rs439401	Heterozygous	Tier3	drug response	5.6	—	7.9	VCV61995
ATIC	c.1503+675T>C	chr2:215347616	rs4673993	Heterozygous	Tier3	drug response	2.1	—	0.8	VCV2259
CHI3L1	p.Arg145Gly	chr1:203183673	rs880633	Homozygous	Tier3	drug response	23.4	0.098	18.5	VCV3900
COMT	c.*522G>A	chr22:19969258	rs165599	Heterozygous	Tier3	drug response	1.8	—	4.5	VCV8289
COMT	c.*764C>T	chr22:19969500	rs165728	Homozygous	Tier3	drug response	1.0	—	0.4	VCV8297
CYP19A1	c.145+418G>T	chr15:51242350	rs727479	Heterozygous	Tier3	drug response	0.0	—	1.1	VCV2581
CYP2C19	c.332-23A>G	chr10:94775367	rs12769205	Homozygous	Tier3	drug response	5.9	—	11.3	VCV6338
CYP3A5	c.219-237A>G	chr7:99672916	rs776746	Homozygous	Tier3	association|drug response|risk factor	3.4	—	20.6	VCV5066
CYP4F2	p.Val433Met	chr19:15879621	rs2108622	Heterozygous	Tier3	drug response	24.9	0.305	13.1	VCV2259
DIO1	c.682-34C>A	chr1:53909897	rs2235544	Heterozygous	Tier3	drug response	2.3	—	3.9	VCV2573
HSD3B1	p.Thr367Asn	chr1:119514623	rs1047303	Homozygous	Tier3	Benign|drug response	0.0	0.126	7.8	VCV1274
ITPA	p.Pro32Thr	chr20:3213196	rs1127354	Heterozygous	Tier2	drug response	23.0	0.312	23.5	VCV1474
MTHFR	p.Ala222Val	chr1:11796321	rs1801133	Heterozygous	Tier2	drug response	27.3	0.842	25.8	VCV3520
NAT2	p.Gly286Glu	chr8:18400860	rs1799931	Heterozygous	Tier2	drug response	0.0	0.022	0.0	VCV725 /
NPC1L1	c.-1473T>C	chr7:44542732	rs2073547	Heterozygous	Tier3	drug response	3.3	—	1.8	VCV18014
OPRM1	p.Asn40Asp	chr6:154039662	rs1799971	Heterozygous	Tier2	Uncertain significance|drug response	24.1	0.104	21.5	VCV9538
OPRM1	c.291-2309C>T	chr6:154087517	rs78937301	Heterozygous	Tier3	drug response	0.0	—	0.2	VCV8284

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessio
OPRM1	c.644-83G>C	chr6:154090869	rs2075572	Heterozygous	Tier3	drug response	0.1	—	0.6	VCV8292
OPRM1	c.1165-11281G>A	chr6:154107402	rs650825	Homozygous	Tier3	drug response	0.2	—	0.0	VCV8283
OPRM1	c.1165-11116A>G	chr6:154107567	rs650245	Homozygous	Tier3	drug response	3.0	—	1.2	VCV8291
OPRM1	c.1165-10726del	chr6:154107951	chr6:154107951	Homozygous	Tier3	drug response	—	—	—	VCV8283
OPRM1	c.1165-9792A>G	chr6:154108891	rs623956	Homozygous	Tier3	drug response	0.9	—	0.4	VCV8291
OPRM1	c.1165-8425C>G	chr6:154110258	rs497332	Homozygous	Tier3	drug response	3.4	—	4.9	VCV8291
OPRM1	c.1165-8253G>A	chr6:154110430	rs606545	Homozygous	Tier3	drug response	5.5	—	4.4	VCV8283
SCN1A	c.603-91G>A	chr2:166053034	rs3812718	Heterozygous	Tier2	drug response	20.4	—	23.1	VCV12897
SLC19A1	p.His27Arg	chr21:45537880	rs1051266	Homozygous	Tier2	drug response	7.5	0.192	9.0	VCV15758
SLC29A3	p.Ser158Phe	chr10:71351651	rs780668	Homozygous	Tier2	Benign|drug response	23.5	0.286	23.6	VCV13034
SLCO1B1	p.Val174Ala	chr12:21178615	rs4149056	Heterozygous	Tier2	drug response	25.1	0.221	24.8	VCV3734
TNF	c.-488G>A	chr6:31575254	rs1800629	Heterozygous	Tier3	drug response	3.1	—	4.8	VCV2259
UGT2B7	c.-161T>C	chr4:69096360	rs7668258	Heterozygous	Tier3	drug response	0.3	—	0.5	VCV8282
UGT2B7	c.-138G>A	chr4:69096383	rs73823859	Heterozygous	Tier3	drug response	0.1	—	14.9	VCV8282
UGT2B7	c.721+201A>C	chr4:69097442	rs12647681	Heterozygous	Tier3	drug response	0.6	—	8.2	VCV8290
UGT2B7	c.721+209A>G	chr4:69097450	rs12647682	Heterozygous	Tier3	drug response	0.2	—	10.0	VCV8290
UGT2B7	c.721+272T>C	chr4:69097513	rs12648462	Heterozygous	Tier3	drug response	4.7	—	5.3	VCV8290

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessio
UGT2B7	`c.722-181C>T`	`chr4:69098359`	`rs73823860`	Heterozygous	Tier3	drug response	0.1	—	0.1	VCV8290
UGT2B7	`c.722-78T>C`	`chr4:69098462`	`rs7439326`	Heterozygous	Tier3	drug response	1.8	—	4.5	VCV8290
UGT2B7	`c.722-49A>G`	`chr4:69098491`	`rs7438244`	Heterozygous	Tier3	drug response	0.5	—	6.7	VCV8282
UGT2B7	`p.Tyr268His`	`chr4:69098620`	`rs7439366`	Heterozygous	Tier3	drug response	6.5	0.066	10.9	VCV8282
UGT2B7	`p.Ala285=`	`chr4:69098673`	`rs34661811`	Heterozygous	Tier3	drug response	4.4	—	18.0	VCV8282
UGT2B7	`p.Leu353=`	`chr4:69107231`	`rs4292394`	Heterozygous	Tier3	drug response	0.1	—	2.3	VCV8282
UGT2B7	`c.1090+64A>T`	`chr4:69107326`	`rs4337789`	Heterozygous	Tier3	drug response	0.1	—	2.6	VCV8282
UGT2B7	`p.Ala397=`	`chr4:69108203`	`rs57913007`	Heterozygous	Tier3	drug response	0.3	—	0.9	VCV8282
UGT2B7	`c.*296T>C`	`chr4:69113032`	`rs6851533`	Heterozygous	Tier3	drug response	6.7	—	4.9	VCV8281
VKORC1	`c.283+837T>C`	`chr16:31092475`	`rs2359612`	Homozygous	Tier2	drug response	0.5	—	12.6	VCV2259
XRCC1	`p.Gln399Arg`	`chr19:43551574`	`rs25487`	Heterozygous	Tier3	drug response	14.4	0.072	8.8	VCV2259

3. Cardiovascular & Channelopathy Modifiers

201 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
AARS2	c.*448G>T	chr6:44300099	rs565803591	Heterozygous	Tier2	Uncertain significance	1.0	—	3.9
ABCG5	p.Ala429Thr	chr2:43823952	rs779125742	Heterozygous	Tier2	Uncertain significance	21.3	0.383	14.3
ABCG8	p.Thr401Ser	chr2:43872296	rs144200355	Heterozygous	Tier2	Conflicting classifications of pathogenicity	13.9	0.083	10.4
ADAMTS10	p.Arg62Gln	chr19:8605262	rs62621197	Heterozygous	Tier3	Benign	23.2	0.018	16.2
APOB	p.Ser4338Asn	chr2:21002409	rs1042034	Homozygous	Tier2	Benign	0.8	0.055	4.3
APOB	p.Thr2515=	chr2:21009323	rs693	Heterozygous	Tier2	Benign	5.4	—	10.5
APOB	p.Ile2313Val	chr2:21009931	rs584542	Homozygous	Tier2	Benign	0.1	0.014	3.6
APOB	p.Asp2312=	chr2:21009932	rs1041968	Heterozygous	Tier2	Benign	0.5	—	7.0
APOB	c.3843-77T>C	chr2:21013610	rs488329	Homozygous	Tier2	Benign	6.3	—	3.1
APOB	p.Leu1060=	chr2:21016593	rs72653073	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	9.1	—	11.5
APOB	p.Thr98Ile	chr2:21041028	rs1367117	Homozygous	Tier2	Benign/Likely benign	22.8	0.093	18.4
APOB	p.Leu12_Leu14del	chr2:21043903	rs17240441	Homozygous	Tier2	Benign/Likely benign	—	—	—
ARID1A	p.Arg1906Gln	chr1:26779615	rs41303631	Heterozygous	Tier2	Benign/Likely benign	20.9	0.125	20.9

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
ATN1	p.Lys236Gln	chr12:6935973	rs200232074	Heterozygous	Tier2	Benign/Likely benign	17.8	0.051	20.1
ATP5F1A	c.-32G>T	chr18:46098263	rs367840441	Heterozygous	Tier3	Benign	17.3	—	20.6
CACNA1A	p.Gln2325dup	chr19:13207859	chr19:13207859	Heterozygous	Tier3	Benign/Likely benign	—	—	—
CACNA1D	c.140-65995C>G	chr3:53431157	rs1156693798	Heterozygous	Tier2	Not reviewed	10.5	—	7.8
CACNA1S	c.4113+7T>C	chr1:201050977	rs6702590	Heterozygous	Tier2	Benign	3.1	—	0.1
CACNA1S	p.Leu522=	chr1:201077934	rs4915476	Homozygous	Tier2	Benign	9.8	—	16.1
CACNA1S	p.Gly517=	chr1:201077947	rs4915477	Homozygous	Tier2	Benign	3.4	—	0.0
CACNA1S	p.Ala69Gly	chr1:201110216	rs12406479	Heterozygous	Tier2	Benign	24.4	0.371	25.0
CACNA1S	c.-231A>G	chr1:201112570	rs4506487	Homozygous	Tier2	Benign	8.8	—	9.7
CASR	p.Asp398Asn	chr3:122262227	rs201177696	Heterozygous	Tier2	Conflicting classifications of pathogenicity	21.8	0.219	14.4
CASZ1	c.-234+4244dup	chr1:10792322	chr1:10792322	Heterozygous	Tier2	Not reviewed	—	—	—
CDKN1C	c.-11+61G>A	chr11:2885573	rs188494894	Heterozygous	Tier2	Conflicting classifications of pathogenicity	15.4	—	16.9
CHST14	p.Gln349Pro	chr15:40472259	rs1463514561	Heterozygous	**Tier1**	Uncertain significance	24.3	0.540	22.9
CHST3	c.*346C>T	chr10:72008817	rs76094341	Heterozygous	Tier2	Uncertain significance	13.5	—	10.5
COL1A1	p.Ala1075Thr	chr17:50188134	rs1800215	Heterozygous	Tier3	Benign/Likely benign	20.9	—	20.2

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
CPLANE1	p.Asp1135Val	chr5:37201694	rs780763762	Heterozygous	Tier2	Not reviewed	19.0	0.173	6.5
CPLANE1	p.Asp1135Tyr	chr5:37201695	rs747635859	Heterozygous	Tier2	Not reviewed	20.8	0.140	7.8
CREBBP	p.Leu551Ile	chr16:3781229	rs61753381	Heterozygous	Tier2	Conflicting classifications of pathogenicity	22.5	0.382	22.4
CRPPA	p.Ser275=	chr7:16301431	rs766874330	Heterozygous	Tier2	Conflicting classifications of pathogenicity	9.9	—	15.8
DCC	c.91+22042A>G	chr18:52362920	rs8095741	Heterozygous	Tier2	Not reviewed	4.1	—	2.7
DHX37	p.Glu267Lys	chr12:124977430	rs33931896	Heterozygous	Tier3	Benign	24.6	0.367	22.3
DMD	c.94-60270G>C	chrX:32910090	rs1923093	Homozygous	Tier2	Not reviewed	1.1	—	5.4
DMD	c.32-76315T>A	chrX:33096515	rs4829276	Heterozygous	Tier2	Not reviewed	1.0	—	0.6
DNAAF1	p.Asp435Asn	chr16:84170131	rs149158199	Heterozygous	Tier3	Benign/Likely benign	8.1	0.071	7.3
DNAAF11	c.1140+8154A>G	chr8:132602012	rs4520144	Heterozygous	Tier2	Not reviewed	1.8	—	1.8
DNAAF2	p.Gly411Arg	chr14:49633919	rs1883243941	Heterozygous	Tier2	Uncertain significance	7.3	0.006	6.1
DNAH1	c.3694-352G>T	chr3:52356262	rs115019178	Heterozygous	Tier2	Not reviewed	11.8	—	19.8
DNAH1	p.Arg3272Cys	chr3:52391251	rs200784189	Heterozygous	Tier2	Uncertain significance	28.4	0.297	25.2
DNAH11	p.Thr400Ile	chr7:21570073	rs72655982	Heterozygous	Tier2	Benign/Likely benign	2.0	0.037	9.8
DNAH11	p.Ala568Thr	chr7:21582013	rs72655988	Heterozygous	Tier3	Benign/Likely benign	21.8	0.132	19.9

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
DNAH11	c.8674-227A>G	chr7:21749451	rs111889929	Heterozygous	Tier2	Likely benign	2.5	—	2.9
DNAH11	p.Ala4315Thr	chr7:21894893	rs72658825	Heterozygous	Tier3	Benign	2.2	0.018	13.0
DYNC2H1	c.366+39T>G	chr11:103113746	rs772875964	Heterozygous	Tier2	Not reviewed	14.7	—	21.9
DYNC2I1	p.Pro923Gln	chr7:158934539	rs73167274	Heterozygous	Tier2	Conflicting classifications of pathogenicity	25.0	0.123	24.0
EBP	c.301+446A>T	chrX:48524518	rs62602501	Heterozygous	Tier2	Not reviewed	0.7	—	0.0
EPAS1	c.*111A>G	chr2:46384771	rs62137032	Heterozygous	Tier3	Benign	15.3	—	13.6
EVC2	p.Arg687His	chr4:5622978	rs144420242	Heterozygous	Tier2	Uncertain significance	16.6	0.262	11.0
FANCA	c.1007-765G>T	chr16:89793312	rs144352354	Heterozygous	Tier2	Not reviewed	0.6	—	8.1
FANCM	p.Asn1253Ser	chr14:45176512	rs45604036	Heterozygous	Tier3	Benign/Likely benign	0.1	0.029	1.4
FANCM	p.Asn1876Ser	chr14:45196458	rs45557033	Heterozygous	Tier3	Benign/Likely benign	1.0	0.019	6.9
FASTKD2	c.*2960T>C	chr2:206794762	rs147381048	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.5	—	2.5
FBN2	p.Ile243Thr	chr5:128464822	rs117524265	Heterozygous	Tier2	Benign/Likely benign	23.3	0.573	20.5
FIG4	c.2546+2077T>G	chr6:109798928	rs4330563	Homozygous	Tier2	Not reviewed	5.1	—	3.9
FIG4	c.2547-9862A>C	chr6:109815226	rs4388322	Homozygous	Tier2	Not reviewed	0.7	—	2.2
FIG4	c.2547-38009T>C	chr6:109839813	rs9400325	Homozygous	Tier2	Not reviewed	0.7	—	0.5
FIG4	c.2547-18770T>C	chr6:109859052	rs9400330	Homozygous	Tier2	Not reviewed	2.3	—	4.4

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
FIG4	c.2547-16414T>G	chr6:109861408	rs9481016	Homozygous	Tier2	Not reviewed	3.2	—	5.2
FRAS1	c.2423-391C>T	chr4:78363122	rs2100484	Heterozygous	Tier2	Not reviewed	2.1	—	1.8
FRAS1	c.2722+2111G>A	chr4:78366165	rs11936250	Heterozygous	Tier2	Not reviewed	0.4	—	2.6
FRAS1	p.Arg3269Gln	chr4:78511299	rs61729366	Heterozygous	Tier2	Benign/Likely benign	32.0	—	24.1
FRAS1	c.11298+469G>A	chr4:78537669	rs7664814	Homozygous	Tier2	Not reviewed	0.4	—	0.8
FREM2	c.*224G>A	chr13:38881011	rs575095989	Heterozygous	Tier2	Uncertain significance	1.5	—	3.6
FREM2	c.*1321G>A	chr13:38882108	rs78527665	Heterozygous	Tier2	Uncertain significance	2.5	—	11.5
GBA1	p.Thr408Met	chr1:155236246	rs75548401	Heterozygous	Tier2	Conflicting classifications of pathogenicity	21.3	0.731	20.0
GJB4	p.Arg124Gln	chr1:34761625	rs140996335	Heterozygous	Tier2	Likely benign	22.1	0.494	16.8
GLI2	p.Arg1526His	chr2:120990542	rs138987487	Heterozygous	Tier2	Benign/Likely benign	29.8	0.256	22.4
GLI3	c.367+28433C>G	chr7:42119793	rs1265401266	Heterozygous	Tier2	Not reviewed	19.4	—	18.4
GPC6	c.877+107375del	chr13:94135255	chr13:94135255	Heterozygous	Tier2	Not reviewed	—	—	—
GTF2IRD2	c.749-149C>T	chr7:74811101	rs71270899	Homozygous	Tier2	Not reviewed	3.5	—	3.7
GUSB	p.Trp627Leu	chr7:65960973	rs1236992554	Heterozygous	**Tier1**	Not reviewed	22.2	0.560	24.9
GYG1	c.7+313G>T	chr3:148991960	rs1000405725	Heterozygous	Tier2	Not reviewed	8.8	—	22.1
HLA-B	p.Arg180Gly	chr6:31356248	rs9266144	Homozygous	Tier2	Not reviewed	0.0	0.251	10.6
HLA-B	p.Leu105Pro	chr6:31356717	rs9266161	Heterozygous	**Tier1**	Not reviewed	22.5	0.205	20.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
HLA-B	p.Leu105Val	chr6:31356718	rs9266162	Heterozygous	Tier2	Not reviewed	8.9	0.188	9.0
HLA-B	p.Ala95Thr	chr6:31356748	rs1131213	Homozygous	Tier2	Not reviewed	13.9	0.179	6.4
HLA-B	p.Gln94His	chr6:31356749	rs1131212	Homozygous	Tier2	Not reviewed	7.1	0.244	11.3
HLA-DRB1	p.Thr262Arg	chr6:32580249	rs9269744	Heterozygous	Tier2	Not reviewed	25.3	0.224	19.7
HLA-DRB1	c.371-270T>G	chr6:32582108	rs28723975	Heterozygous	Tier2	Not reviewed	6.4	—	11.3
HLA-DRB1	c.371-297A>G	chr6:32582135	rs28723976	Heterozygous	**Tier1**	Not reviewed	7.4	—	12.9
HLA-DRB1	c.371-396T>C	chr6:32582234	rs28723982	Heterozygous	Tier2	Not reviewed	0.4	—	2.1
HLA-DRB1	c.371-415G>A	chr6:32582253	rs28723984	Heterozygous	Tier2	Not reviewed	5.1	—	10.3
HLA-DRB1	p.Tyr107Ter	chr6:32584158	rs11554463	Heterozygous	Tier2	Not reviewed	28.0	—	27.2
HLA-DRB1	p.Tyr107Phe	chr6:32584159	rs17879620	Heterozygous	Tier2	Not reviewed	0.1	0.146	0.0
HLA-DRB1	p.Tyr107Asp	chr6:32584160	rs17884043	Heterozygous	Tier2	Not reviewed	16.5	0.132	8.6
HLA-DRB1	p.Ala103Pro	chr6:32584172	rs16822805	Heterozygous	**Tier1**	Uncertain significance	0.0	0.209	20.2
HLA-DRB1	p.Ala102Gly	chr6:32584174	rs17878857	Heterozygous	Tier2	Not reviewed	12.2	0.311	11.4
HLA-DRB1	p.Ala102GlyfsTer26	chr6:32584175	chr6:32584175	Heterozygous	Tier2	Not reviewed	—	—	—
HLA-DRB1	p.Gln99His	chr6:32584182	rs17879599	Heterozygous	Tier2	Not reviewed	0.6	0.217	11.3
HLA-DRB1	p.Gln99Glu	chr6:32584184	rs17881965	Heterozygous	Tier2	Not reviewed	0.0	0.281	2.3
HLA-DRB1	p.Pro40Ser	chr6:32584361	rs9269955	Heterozygous	Tier2	Not reviewed	0.0	0.034	0.0
HLA-DRB1	p.Pro40Ala	chr6:32584361	rs9269955	Heterozygous	Tier2	Not reviewed	0.0	0.028	0.2
HLA-DRB1	p.Gln39Ter	chr6:32584364	rs9269957	Heterozygous	Tier2	Uncertain significance	26.8	—	28.2
HLA-DRB1	c.101-43C>T	chr6:32584421	rs28724109	Heterozygous	Tier2	Not reviewed	6.7	—	19.2

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
HLA-DRB1	p.Ala14Val	chr6:32589702	rs9270302	Heterozygous	Tier2	Not reviewed	15.4	0.177	12.1
HSPG2	c.9710-10G>A	chr1:21839560	rs369434104	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.4	—	0.6
HSPG2	p.Arg257Trp	chr1:21887609	rs535956121	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.3	0.203	0.8
HYDIN	p.Arg4513His	chr16:70834028	rs201554059	Heterozygous	Tier2	Likely benign	22.4	0.213	18.0
HYDIN	p.Thr690Ala	chr16:71067297	rs10744982	Heterozygous	Tier2	Not reviewed	24.9	0.082	21.7
IKBKG	c.1056-6C>T	chrX:154563953	rs5945206	Homozygous	Tier2	Uncertain significance	10.5	—	7.3
IQSEC2	p.Ala1470Thr	chrX:53234278	rs191886831	Heterozygous	Tier3	Benign	14.2	0.028	14.8
KDM6A	c.384+5299del	chrX:44980000	chrX:44980000	Heterozygous	Tier2	Not reviewed	—	—	—
KIF1B	p.Ser1427Asn	chr1:10361801	rs755113137	Heterozygous	Tier2	Uncertain significance	25.9	0.236	24.2
KIF20A	p.Pro839Leu	chr5:138187256	rs3172747	Heterozygous	Tier3	Benign	23.9	0.163	22.9
KLHL24	c.-62+617T>C	chr3:183644159	rs9877526	Homozygous	Tier2	Not reviewed	6.0	—	0.7
KMT2C	c.4661-778C>A	chr7:152188625	rs13242182	Heterozygous	Tier2	Not reviewed	1.1	—	0.2
LDLRAP1	p.Ile132=	chr1:25557204	rs369497122	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.2	—	0.4
LMBRD1	p.Leu135fs	chr6:69749975	chr6:69749975	Heterozygous	Tier2	Not reviewed	—	—	—
MAPRE2	p.Arg8fs	chr18:34979998	chr18:34979998	Heterozygous	Tier2	Not reviewed	—	—	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
MC2R	c.*992A>T	chr18:13883633	rs897474751	Heterozygous	Tier2	Uncertain significance	4.4	—	0.1
MRPS22	c.-71-65596T>G	chr3:139278360	rs78362422	Heterozygous	Tier2	Not reviewed	2.5	—	1.3
MYH11	c.-18+2381G>C	chr16:15854560	rs9933232	Heterozygous	Tier2	Not reviewed	0.5	—	1.6
MYL2	c.-52-2913T>C	chr12:110923494	rs12828640	Heterozygous	Tier2	Not reviewed	5.4	—	4.6
MYL2	c.-52-15019A>G	chr12:110935600	rs4426183	Heterozygous	Tier2	Not reviewed	0.5	—	0.3
MYO18B	c.6157-7087G>T	chr22:25985276	rs9306420	Homozygous	Tier2	Not reviewed	13.1	—	1.5
MYOT	c.-233C>A	chr5:137867932	rs186433387	Heterozygous	Tier3	Benign	16.3	—	18.8
NDUFS2	p.Arg323Gln	chr1:161210692	rs35086265	Heterozygous	Tier2	Conflicting classifications of pathogenicity	16.8	0.525	8.7
NEB	c.*201A>T	chr2:151485559	rs190132249	Heterozygous	Tier2	Conflicting classifications of pathogenicity	1.9	—	3.7
NEB	p.Ala4223Thr	chr2:151606686	rs12998234	Homozygous	Tier2	Benign/Likely benign	8.2	0.045	9.3
NEB	p.Glu191Gln	chr2:151724301	rs35686968	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.0	0.324	25.3
NF1	c.*126G>A	chr17:31374281	rs184715593	Heterozygous	Tier2	Conflicting classifications of pathogenicity	1.9	—	6.9
NFIX	c.27+6825_27+6853del	chr19:13002688	chr19:13002688	Heterozygous	Tier2	Not reviewed	—	—	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
NFIX	`c.559+1309_559+1310del`	`chr19:13026858`	`rs1169659113`	Heterozygous	Tier2	Not reviewed	—	—	—
NFIX	`c.1255-191dup`	`chr19:13087799`	`chr19:13087799`	Homozygous	Tier2	Not reviewed	14.4	—	—
NIPBL	`c.-79-1552_-79-1545del`	`chr5:36952054`	`chr5:36952054`	Heterozygous	Tier3	Benign	—	—	—
NOTCH1	`p.Arg1279His`	`chr9:136506781`	`rs61751543`	Heterozygous	Tier3	Benign/Likely benign	22.5	0.098	14.1
NSUN2	`p.Val715Ile`	`chr5:6600087`	`rs112951498`	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.0	0.011	11.0
PCSK9	`c.207+15A>G`	`chr1:55040059`	`rs2495482`	Homozygous	Tier2	Benign	8.7	—	3.8
PCSK9	`c.658-7C>T`	`chr1:55052643`	`rs2483205`	Homozygous	Tier2	Benign	1.3	—	0.3
PCSK9	`c.799+3A>G`	`chr1:55052794`	`rs2495477`	Homozygous	Tier2	Benign	14.3	—	13.7
PCSK9	`p.Gln342=`	`chr1:55057360`	`rs509504`	Homozygous	Tier2	Conflicting classifications of pathogenicity	8.6	—	13.3
PCSK9	`p.Val460=`	`chr1:55058524`	`rs540796`	Heterozygous	Tier2	Benign	8.8	—	15.2
PCSK9	`p.Val474Ile`	`chr1:55058564`	`rs562556`	Heterozygous	Tier2	Benign/Likely benign	0.0	0.051	0.2
PCSK9	`p.Gly670Glu`	`chr1:55063514`	`rs505151`	Homozygous	Tier2	Benign/Likely benign	1.7	0.067	12.1
PCSK9	`c.*571C>T`	`chr1:55064155`	`rs662145`	Heterozygous	Tier2	Benign	0.7	—	0.3
PEX10	`p.Thr274Ala`	`chr1:2406576`	`rs34154371`	Heterozygous	Tier3	Benign/Likely benign	23.6	0.357	18.0
PEX13	`p.Leu360Val`	`chr2:61048636`	`rs74350038`	Heterozygous	Tier3	Benign/Likely benign	1.8	0.151	11.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
PEX16	p.Val254Leu	chr11:45914138	rs35214605	Heterozygous	Tier3	Benign/Likely benign	15.4	0.046	18.8
PIBF1	p.His523Asn	chr13:72908609	rs115858628	Heterozygous	Tier3	Benign/Likely benign	19.6	0.085	19.7
PKP2	p.Val543Ile	chr12:32824092	rs146102241	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.2	0.333	22.9
PKP2	p.Ser70Ile	chr12:32896523	rs75909145	Heterozygous	Tier3	Benign/Likely benign	21.6	0.155	14.8
PLEC	p.Arg296Gln	chr8:143934868	rs138924815	Heterozygous	Tier3	Benign	24.6	0.398	22.0
PNPLA2	p.Pro265Ser	chr11:823729	rs56152088	Heterozygous	Tier3	Benign	13.9	0.111	19.6
POGZ	c.284-4376T>C	chr1:151435217	rs11578888	Homozygous	Tier2	Not reviewed	0.1	—	0.5
POLR3A	c.*620G>A	chr10:77976858	rs184043149	Heterozygous	Tier2	Uncertain significance	0.0	—	0.0
POMT1	p.Asp411Glu	chr9:131515483	rs11243406	Heterozygous	Tier3	Benign	23.9	0.464	24.3
RASA1	p.Ala99Val	chr5:87268747	rs111840875	Heterozygous	Tier3	Benign/Likely benign	15.9	0.189	16.6
RYR1	p.Leu198=	chr19:38444640	rs2229139	Heterozygous	Tier2	Benign/Likely benign	10.4	—	12.1
RYR1	p.Ala359=	chr19:38448768	rs10406027	Homozygous	Tier2	Benign	0.4	—	0.0
RYR1	p.Asn993=	chr19:38466199	rs2228070	Heterozygous	Tier2	Benign	0.9	—	0.0
RYR1	p.Ile1152=	chr19:38469040	rs11083462	Heterozygous	Tier2	Benign	5.6	—	0.0
RYR1	p.Ala2500=	chr19:38500876	rs2228072	Heterozygous	Tier2	Benign	0.8	—	0.0
RYR1	p.Val2509=	chr19:38500903	rs2071088	Heterozygous	Tier2	Benign	4.2	—	11.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
RYR1	c.7835+5A>G	chr19:38502732	rs1469695	Heterozygous	Tier2	Benign	2.2	—	9.1
RYR1	p.His2621=	chr19:38502907	rs2229142	Heterozygous	Tier2	Benign	6.4	—	11.9
RYR1	p.Thr2659=	chr19:38504270	rs2229144	Heterozygous	Tier2	Benign	0.5	—	0.0
RYR1	p.Ile2706=	chr19:38504798	rs2960340	Heterozygous	Tier2	Benign	0.8	—	0.2
RYR1	p.Asp2730=	chr19:38504870	rs2915951	Heterozygous	Tier2	Benign	0.3	—	0.0
RYR1	p.Glu2779=	chr19:38505335	rs2915952	Heterozygous	Tier2	Benign	6.4	—	12.9
RYR1	p.Ser2863=	chr19:38506350	rs2229146	Heterozygous	Tier2	Benign	3.7	—	0.1
RYR1	c.8693-10G>C	chr19:38506819	rs2915958	Heterozygous	Tier2	Benign	5.2	—	4.1
RYR1	p.Pro3062=	chr19:38512085	rs2071089	Heterozygous	Tier2	Benign	0.1	—	0.0
RYR2	p.Ser1400Gly	chr1:237591776	rs56229512	Heterozygous	Tier3	Benign/Likely benign	19.2	0.310	17.5
SCAPER	c.1866+8A>G	chr15:76753800	rs139187840	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.4	—	7.1
SCN4A	c.*1069G>A	chr17:63939702	rs368616841	Heterozygous	Tier2	Uncertain significance	0.0	—	0.0
SGCD	c.-44+62149G>C	chr5:156186168	rs564618852	Heterozygous	Tier2	Not reviewed	6.7	—	2.6
SIX3	c.-1863G>A	chr2:44940242	rs1353919432	Heterozygous	Tier2	Not reviewed	20.2	—	19.3
SMARCA4	c.4425-5369C>T	chr19:11052886	rs1529729	Heterozygous	Tier2	Not reviewed	0.6	—	1.7
SMARCA4	c.*327T>C	chr19:11062143	rs146968124	Heterozygous	Tier3	Benign	15.2	—	12.5
SMO	p.Pro698Arg	chr7:129212180	rs116640950	Heterozygous	Tier2	Likely benign	12.4	0.153	19.9
SMPX	c.*14+6246G>C	chrX:21731303	rs1015473	Heterozygous	Tier2	Not reviewed	1.1	—	9.4

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
SON	p.Tyr1230_Thr1247dup	chr21:33552918	rs1569055881	Heterozygous	Tier3	Likely benign	—	—	—
SPEG	p.Pro674Arg	chr2:219449179	rs757668535	Heterozygous	Tier2	Uncertain significance	12.8	0.050	20.7
SPEN	p.Glu1517Val	chr1:15930790	rs143050516	Heterozygous	Tier2	Likely benign	25.3	0.109	22.7
SPTBN1	c.-48+1068G>T	chr2:54457586	rs995589477	Heterozygous	Tier2	Not reviewed	20.5	—	20.4
STIL	c.*545C>T	chr1:47250591	rs181264069	Heterozygous	Tier2	Uncertain significance	1.4	—	0.7
STRA6	p.Arg408Gln	chr15:74183933	rs150814749	Heterozygous	Tier2	Benign/Likely benign	0.6	0.159	16.5
SYNE2	c.18038+4450A>G	chr14:64194687	rs748933084	Heterozygous	Tier2	Not reviewed	9.4	—	9.0
TARS1	c.330-4del	chr5:33453265	chr5:33453265	Heterozygous	Tier2	Not reviewed	—	—	—
THSD4	c.1153-113066G>A	chr15:71547464	rs117286812	Heterozygous	Tier3	Benign	18.2	0.049	23.8
TMEM107	c.*622G>A	chr17:8173581	rs201266955	Heterozygous	Tier2	Conflicting classifications of pathogenicity	14.5	—	22.9
TMEM43	p.Trp316Ser	chr3:14139244	rs199526104	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	32.0	0.533	26.5
TMEM43	c.*115T>C	chr3:14141910	rs1043943	Heterozygous	Tier2	Benign/Likely benign	1.6	—	9.2
TRIP11	p.Arg1576His	chr14:91999405	rs35007347	Heterozygous	Tier3	Benign/Likely benign	23.4	0.137	22.0
TTN	p.Asn16662Thr	chr2:178612540	rs36043230	Heterozygous	Tier3	Benign/Likely benign	23.3	0.328	22.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)
TTN	`p.Pro10462Thr`	`chr2:178694641`	`rs761102022`	Heterozygous	Tier2	Uncertain significance	20.1	0.381	22.6
TTR	`c.-1054C>G`	`chr18:31556179`	`chr18:31556179`	Heterozygous	Tier2	Not reviewed	8.6	—	15.1
VCL	`p.His27=`	`chr10:73998288`	`rs200733607`	Heterozygous	Tier2	Conflicting classifications of pathogenicity	13.7	—	16.0
VCL	`c.3153+47T>C`	`chr10:74114434`	`rs2270550`	Homozygous	Tier2	Benign	0.3	—	5.1
VPS13A	`p.Gln2640=`	`chr9:77357805`	`rs200387635`	Heterozygous	Tier2	Conflicting classifications of pathogenicity	5.4	—	0.3
WAC	`c.274+13301C>G`	`chr10:28549058`	`rs913821`	Heterozygous	Tier2	Not reviewed	0.1	—	0.5
WARS2	`p.His151Arg`	`chr1:119042327`	`rs150022801`	Heterozygous	Tier2	Likely benign	14.3	0.062	17.8
WFS1	`c.-5-2548G>C`	`chr4:6274903`	`rs62283056`	Heterozygous	Tier2	Not reviewed	3.4	—	7.9
WRN	`c.2967+88G>A`	`chr8:31132594`	`rs2252480`	Heterozygous	Tier2	Benign	1.3	—	0.7
ZEB2	`c.74-11140_74-11139dup`	`chr2:144441167`	`chr2:144441167`	Heterozygous	Tier2	Not reviewed	—	—	—
ZEB2	`c.-69-197A>T`	`chr2:144517616`	`rs533595036`	Heterozygous	Tier2	Not reviewed	20.7	—	18.6
ZEB2	`c.-69-198C>G`	`chr2:144517617`	`rs558617021`	Heterozygous	Tier2	Not reviewed	17.8	—	15.7
ZFPM2	`c.420+34535_420+34536del`	`chr8:105596005`	`chr8:105596005`	Heterozygous	Tier2	Not reviewed	—	—	—
ZFPM2	`c.533-57576C>T`	`chr8:105731142`	`rs544662446`	Heterozygous	Tier2	Not reviewed	19.7	—	14.7

4. Metabolic, Mitochondrial & DNA Repair Machinery

0 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessions

5. Protective & Longevity Modifiers

9 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Extern Acces
ADH1C	p.Ile350Val	chr4:99339632	rs698	Heterozygous	Tier2	protective	15.4	—	15.9	VCV18
ADH1C	p.Arg272Gln	chr4:99342808	rs1693482	Heterozygous	Tier2	protective	10.4	—	12.8	VCV18
CASP8	c.-937_-932del	chr2:201232809	rs3834129	Heterozygous	Tier3	protective	—	—	—	VCV77
CD209	c.-336T>C	chr19:7747847	rs4804803	Heterozygous	Tier3	protective|risk factor	3.8	—	1.9	VCV54
EDNRA	c.-230G>A	chr4:147481217	rs1801708	Heterozygous	Tier3	protective	16.2	—	13.5	VCV16
FCGR2B	p.Ile232Thr	chr1:161674008	rs1050501	Homozygous	Tier3	protective|risk factor	10.9	0.024	11.2	VCV54
GNPTAB	c.117+9523G>T	chr12:101821036	rs11111044	Heterozygous	Tier2	Not reviewed	0.2	—	0.4	— / At
GNPTAB	c.117+3554G>C	chr12:101827005	rs7964859	Heterozygous	Tier2	Not reviewed	0.7	—	0.6	— / At
TLR1	p.Ser602Ile	chr4:38797027	rs5743618	Heterozygous	Tier2	Conflicting classifications of pathogenicity|protective	7.5	0.056	7.9	VCV83

6. Secondary & Exploratory Clinical Variants

556 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
ABCA12	p.Lys982Thr	chr2:215000939	rs142196906	Heterozygous	Tier2	Conflicting classifications of pathogenicity	18.4	0.354
ABCC4	c.3871-6349C>T	chr13:95028031	rs4148546	Heterozygous	Tier2	Not reviewed	1.9	—
ABCD3	p.Ala131Thr	chr1:94473821	rs138378383	Heterozygous	Tier2	Uncertain significance	22.9	0.492
ACAA2	p.Val253Ile	chr18:49791596	rs377532103	Heterozygous	Tier2	Likely benign	1.4	0.307
ACADSB	c.*3332A>T	chr10:123057097	rs763405920	Heterozygous	Tier2	Uncertain significance	10.4	—
ACHE	p.His353Asn	chr7:100893176	rs1799805	Heterozygous	Tier3	Benign	19.0	0.038
ACOX3	c.-15+600A>G	chr4:8440048	rs62287361	Heterozygous	**Tier1**	Not reviewed	3.3	—
ACSF3	p.Arg39Cys	chr16:89100796	rs779775679	Heterozygous	Tier2	Uncertain significance	4.0	0.038
ACSM2B	p.Tyr540Asn	chr16:20540665	rs139253186	Heterozygous	**Tier1**	Not reviewed	27.6	0.401
ACTN4	p.Asp581Asn	chr19:38724205	rs768314562	Heterozygous	Tier2	Not reviewed	21.6	0.179
ACTR5	p.Met148Ile	chr20:38750078	rs34862565	Heterozygous	Tier3	Not reviewed	23.7	0.757
ADAMTS18	p.Pro12Arg	chr16:77434661	rs200952997	Heterozygous	Tier2	Benign/Likely benign	15.9	0.033
ADAMTS7	c.1029-221G>A	chr15:78790059	rs142754252	Heterozygous	**Tier1**	Not reviewed	0.2	—
ADAMTSL4	p.Pro381Arg	chr1:150554375	rs145639662	Heterozygous	Tier2	Conflicting classifications of pathogenicity	22.4	0.037
ADCY10	p.Pro1353Leu	chr1:167823118	rs72697797	Heterozygous	Tier3	Benign/Likely benign	0.7	0.128

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
ADCY8	c.1111-10720C>T	chr8:130962718	rs36109123	Heterozygous	Tier2	Not reviewed	4.9	—
ADGRV1	p.Ala2803Thr	chr5:90705420	rs111033530	Heterozygous	Tier3	Benign/Likely benign	25.6	0.215
ADGRV1	p.Thr4090Asn	chr5:90763453	rs199839743	Heterozygous	Tier2	Benign/Likely benign	22.6	0.211
AEBP1	p.Ala1031Thr	chr7:44113875	rs146344486	Heterozygous	Tier2	Benign/Likely benign	23.2	0.258
AEBP1	p.Val1148Ile	chr7:44114226	rs13898	Heterozygous	Tier3	Benign/Likely benign	20.9	0.221
AFG2B	c.2094+15del	chr15:45419998	chr15:45419998	Heterozygous	Tier2	Not reviewed	—	—
AGT	p.Ala286Ser	chr1:230706174	rs377397992	Heterozygous	Tier2	Uncertain significance	1.3	0.133
AGTR1	c.-47-8809C>T	chr3:148732180	rs146439726	Heterozygous	Tier2	Not reviewed	7.8	—
AIFM3	p.Arg401Trp	chr22:20976921	rs1365055212	Heterozygous	Tier2	Not reviewed	29.2	0.277
AIMP1	p.Glu120Gly	chr4:106328211	rs367813302	Heterozygous	Tier2	Uncertain significance	20.3	0.100
AIRE	c.1279-18C>T	chr21:44293771	rs72650678	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.8	—
AK4P3	p.Ala177Val	chr12:31615910	rs35037649	Homozygous	Tier2	Not reviewed	7.1	—
AKAP1	p.Ser330Gly	chr17:57106452	rs34535433	Heterozygous	Tier3	Benign	0.1	0.021
AKAP5	p.Ser205Phe	chr14:64469008	rs2230492	Heterozygous	Tier2	Not reviewed	23.1	0.176
ALOXE3	c.*514T>G	chr17:8096113	rs80223722	Heterozygous	Tier2	Uncertain significance	13.1	—
ALPK1	c.-101+27968G>A	chr4:112343820	chr4:112343820	Heterozygous	Tier2	Not reviewed	17.2	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
ALPL	p.Leu252=	chr1:21568211	rs867145238	Heterozygous	Tier2	Conflicting classifications of pathogenicity	9.6	—
AMPD1	p.Glu435Gly	chr1:114677435	rs764950035	Heterozygous	**Tier1**	Uncertain significance	32.0	0.949
ANK2	c.85-22668G>A	chr4:113151748	rs11929834	Heterozygous	Tier2	Not reviewed	8.0	—
ANO1	p.Arg147fs	chr11:70089999	chr11:70089999	Heterozygous	Tier2	Not reviewed	—	—
APOL1	p.Leu7Val	chr22:36254974	rs372661836	Heterozygous	Tier2	Uncertain significance	14.6	0.044
APOO	c.389-16A>G	chrX:23858749	rs373181632	Heterozygous	Tier2	Not reviewed	14.2	—
AR	p.Gln67_Gln80del	chrX:67545317	chrX:67545317	Heterozygous	Tier3	Likely benign	—	—
AR	p.Gly461_Gly473del	chrX:67546515	chrX:67546515	Heterozygous	Tier3	Benign	—	—
ARG1	c.306-388A>G	chr6:131580831	rs779800455	Heterozygous	Tier2	Not reviewed	7.6	—
ARHGEF2	p.Ser20ProfsTer?	chr1:156006982	rs558141032	Heterozygous	Tier2	Not reviewed	—	—
ASAH2	p.Ala623Ser	chr10:50196910	rs1444286757	Heterozygous	Tier2	Not reviewed	19.1	0.108
ASB3	c.1238+1126G>A	chr2:53699145	rs72793655	Heterozygous	Tier2	Not reviewed	0.9	—
ASXL3	c.54+3289C>T	chr18:33581974	rs538726349	Heterozygous	Tier2	Not reviewed	18.7	—
ATG4B	p.Pro188Ala	chr2:241666668	rs201601658	Heterozygous	Tier2	Not reviewed	8.8	0.023
ATP1A1	p.Ile580Val	chr1:116395187	rs77217304	Heterozygous	Tier3	Benign	1.1	0.053
ATP6V0A2	p.Asn850Ser	chr12:123758010	rs75279884	Heterozygous	Tier3	Benign	0.0	0.191
ATP6V0A4	c.-29G>A	chr7:138786169	rs145258624	Heterozygous	Tier2	Uncertain significance	7.2	—
ATP8A2	c.2384+2419_2384+2422dup	chr13:25701726	chr13:25701726	Heterozygous	Tier2	Not reviewed	—	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
ATR	p.Val959Met	chr3:142550233	rs28910271	Heterozygous	Tier3	Benign	12.7	0.065
ATR	p.Val316Ile	chr3:142562456	rs28897764	Heterozygous	Tier3	Benign	17.3	0.034
ATXN1L	p.Pro597His	chr16:71851530	rs61747556	Heterozygous	Tier2	Not reviewed	19.3	0.058
ATXN2	p.Gln25_Gln28dup	chr12:111598964	rs1555246688	Heterozygous	Tier3	Benign	—	—
AWAT2	p.Ile96Val	chrX:70043664	rs61735680	Heterozygous	Tier2	Not reviewed	16.2	0.126
BACH2	c.-12-16333C>T	chr6:90025189	rs4388268	Heterozygous	Tier2	Not reviewed	4.6	—
BCAS3	c.1487-5702C>T	chr17:61010049	rs571142153	Heterozygous	Tier2	Not reviewed	17.4	—
BCKDHB	c.951+20710T>G	chr6:80223922	chr6:80223922	Heterozygous	Tier2	Not reviewed	16.6	—
BCL11A	c.386-8965T>A	chr2:60477798	rs7593947	Homozygous	Tier2	Not reviewed	2.0	—
BCL11B	c.58+3615_58+3616dup	chr14:99267548	chr14:99267548	Heterozygous	Tier2	Not reviewed	—	—
BLTP1	p.Pro802Ala	chr4:122219496	rs1048893554	Heterozygous	Tier2	Uncertain significance	23.7	0.409
BMP8B	c.674-430G>C	chr1:39765247	rs183659449	Heterozygous	Tier2	Not reviewed	6.1	—
BMP8B	p.Gly92Cys	chr1:39788212	rs769660218	Heterozygous	Tier2	Uncertain significance	22.7	0.159
BNC2	c.3+1897C>T	chr9:16868749	rs1053680531	Heterozygous	Tier2	Not reviewed	16.7	—
BPTF	c.1865-5726A>G	chr17:67886118	rs117415449	Heterozygous	Tier3	Benign	21.7	—
BSN	p.Ser3307Gly	chr3:49661764	rs145931220	Heterozygous	Tier2	Likely benign	1.8	0.075
C10orf67	p.Ala524fs	chr10:23209999	chr10:23209999	Heterozygous	Tier2	Not reviewed	—	—
C10orf67	c.1571-5745_1571-5744del	chr10:23210001	chr10:23210001	Heterozygous	Tier2	Not reviewed	—	—
C4A	p.Ser347Tyr	chr6:31984403	rs392610	Heterozygous	Tier2	Benign	26.5	0.212
C4A	p.Arg477Trp	chr6:31991819	rs1554297700	Heterozygous	Tier2	Not reviewed	15.9	0.183

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C4B	p.Thr907Ala	chr6:32027362	rs796750528	Heterozygous	Tier2	Benign/Likely benign	5.9	0.034
CA8	c.418-13115A>G	chr8:60245494	rs9643367	Heterozygous	Tier2	Not reviewed	16.8	—
CACNA1E	c.3829-920G>A	chr1:181754317	rs147545823	Homozygous	Tier2	Not reviewed	10.6	—
CACNA1F	p.Arg508Gln	chrX:49226037	rs34162630	Heterozygous	Tier3	Benign/Likely benign	25.9	0.715
CACNA1G	c.354+31_354+34del	chr17:50568986	chr17:50568986	Heterozygous	Tier2	Conflicting classifications of pathogenicity	—	—
CACNA1H	c.1120-3C>T	chr16:1200713	rs559931034	Heterozygous	Tier2	Uncertain significance	5.8	—
CACNA1H	p.Arg2218His	chr16:1220585	rs56885166	Heterozygous	Tier2	Benign/Likely benign	25.6	0.282
CACNA2D1	c.294+36117T>C	chr7:82299018	rs6969265	Heterozygous	Tier2	Not reviewed	1.1	—
CANT1	p.Arg19Gln	chr17:78997567	rs144060377	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.5	0.559
CAPN2	p.Glu521Gln	chr1:223761612	rs28370127	Heterozygous	Tier2	Not reviewed	21.8	0.219
CARD10	p.Arg601Gln	chr22:37497164	rs201423754	Heterozygous	Tier2	Uncertain significance	10.7	0.070
CARMIL2	p.Lys937del	chr16:67652329	rs144268948	Heterozygous	Tier3	Benign	—	—
CARS2	p.Gly216Glu	chr13:110683059	rs146773721	Heterozygous	Tier3	Benign	1.5	0.060
CATSPER2	c.-159+765G>A	chr15:43663653	rs1555451673	Heterozygous	Tier3	Not reviewed	18.4	—
CATSPER2	c.-248T>C	chr15:43664507	rs1306890410	Heterozygous	Tier2	Not reviewed	21.1	—
CATSPER2	c.-285+323G>A	chr15:43667705	rs143039626	Homozygous	Tier2	Not reviewed	20.3	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
CCDC80	p.Tyr938His	chr3:112605458	rs114697626	Heterozygous	Tier2	Likely benign	26.5	0.235
CD109	p.Lys789Glu	chr6:73787261	rs45572237	Heterozygous	Tier2	Not reviewed	8.5	0.045
CD177	c.620-1440T>G	chr19:43358825	rs1568444200	Homozygous	Tier2	Not reviewed	2.2	—
CD177	c.620-1117A>G	chr19:43359148	rs71339505	Homozygous	Tier2	Not reviewed	6.3	—
CD177	c.620-1074T>C	chr19:43359191	rs71339506	Homozygous	Tier2	Not reviewed	7.5	—
CD177	c.620-1022T>C	chr19:43359243	rs755694960	Homozygous	Tier2	Not reviewed	2.2	—
CD40	c.559+36G>C	chr20:46126737	rs41282788	Heterozygous	**Tier1**	Benign	2.0	—
CDAN1	p.Val328Ile	chr15:42735336	rs144448301	Heterozygous	Tier2	Uncertain significance	17.1	0.237
CDH1	c.1711+343C>T	chr16:68819768	rs4783689	Homozygous	Tier2	Not reviewed	0.6	—
CDH12	c.527-17594G>A	chr5:21872384	rs1032542229	Heterozygous	Tier2	Not reviewed	17.1	—
CDH8	c.1537-6038_1537-6037del	chr16:61719997	chr16:61719997	Heterozygous	Tier2	Not reviewed	—	—
CDHR1	p.His53Gln	chr10:84196512	rs12781048	Heterozygous	Tier3	Benign	0.0	0.144
CEBPE	p.Leu155Met	chr14:23118629	rs141903485	Heterozygous	Tier2	Benign/Likely benign	25.6	0.113
CELA1	p.Tyr5Ser	chr12:51346625	rs117443541	Homozygous	Tier2	Not reviewed	14.8	0.171
CELA1	p.Tyr5His	chr12:51346626	rs116944010	Homozygous	Tier2	Not reviewed	15.2	0.206
CELA1	p.Leu4AlafsTer21	chr12:51346630	rs377599213	Homozygous	Tier2	Benign	0.8	—
CELA1	p.Val3ProfsTer21	chr12:51346631	rs1491233894	Homozygous	Tier2	Benign	—	—
CEP63	p.Gln185His	chr3:134537268	rs114108011	Heterozygous	**Tier1**	Benign	34.0	0.276
CEP85L	c.1020+5170C>T	chr6:118560359	rs537800022	Heterozygous	Tier2	Uncertain significance	0.2	—

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CERS2	p.Val267Ile	chr1:150966805	rs149992226	Heterozygous	Tier2	Likely benign	0.1	0.259
CES1	p.Arg4Pro	chr16:55833045	rs111604615	Heterozygous	Tier2	Not reviewed	8.5	0.145
CHD2	c.1153+109G>C	chr15:92944624	rs1320819	Heterozygous	Tier2	Benign	0.2	—
CHD2	c.1502+505G>C	chr15:92949581	rs4327000	Heterozygous	Tier2	Not reviewed	1.4	—
CHODL	c.-45+109054_-45+109055insGAA	chr21:18026454	chr21:18026454	Homozygous	Tier2	Not reviewed	—	—
CHRD	p.Pro10Ser	chr3:184380346	chr3:184380346	Heterozygous	Tier2	Not reviewed	21.7	0.015
CHRNE	p.Gly18Val	chr17:4902757	rs4790235	Heterozygous	Tier3	Benign	7.0	0.128
CLEC4M	p.Arg187Gln	chr19:7765983	rs28413926	Homozygous	Tier2	Not reviewed	0.0	0.027
CLEC4M	p.Glu210ArgfsTer3	chr19:7766049	rs2034253170	Homozygous	Tier2	Not reviewed	—	—
CLEC4M	p.Glu210AlafsTer7	chr19:7766051	rs867796501	Homozygous	Tier2	Not reviewed	—	—
CNGB1	c.159+14C>T	chr16:57967114	rs199591689	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.0	—
CNTN6	c.762-5763G>A	chr3:1315887	rs155231	Homozygous	Tier2	Not reviewed	0.2	—
CNTNAP2	c.*3999del	chr7:148419601	chr7:148419601	Heterozygous	Tier2	Uncertain significance	—	—
CNTNAP3C	c.85+1G>A	chr9:61331053	rs71506495	Heterozygous	Tier2	Not reviewed	25.8	—
CNTNAP5	c.2077+37493_2077+37494insA	chr2:124685452	rs201059908	Heterozygous	Tier2	Not reviewed	20.2	—
CNTNAP5	c.2077+37494T>A	chr2:124685452	rs149834795	Heterozygous	Tier2	Not reviewed	20.2	—
CNTNAP5	c.2077+37495T>A	chr2:124685453	rs142344379	Heterozygous	Tier2	Not reviewed	21.6	—
CNTNAP5	p.Pro715Arg	chr2:124747295	rs200759902	Heterozygous	Tier2	Not reviewed	13.9	0.018
COG1	p.Ala20Pro	chr17:73193127	rs142719529	Heterozygous	Tier3	Benign/Likely benign	29.8	0.168

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
COG5	c.1476-14T>G	chr7:107281413	rs769339809	Heterozygous	Tier3	Likely benign	17.0	—
COG5	c.538+54830T>C	chr7:107472407	chr7:107472407	Heterozygous	Tier2	Not reviewed	18.9	—
COL15A1	p.Phe851Leu	chr9:99042084	rs35901514	Heterozygous	Tier3	Benign	23.9	0.412
COL23A1	c.361+42073T>G	chr5:178518609	rs1350094006	Heterozygous	Tier2	Not reviewed	7.6	—
COL28A1	p.Gly1066Glu	chr7:7360398	rs200146803	Heterozygous	Tier2	Uncertain significance	1.4	0.110
COL4A1	c.1286-281T>C	chr13:110195399	rs12184772	Heterozygous	Tier3	Benign	0.0	—
COL4A2	c.4882-83T>C	chr13:110511851	rs2479426	Heterozygous	Tier3	Benign	1.5	—
COL4A4	c.72-854C>T	chr2:227145412	rs2001851	Heterozygous	Tier2	Not reviewed	0.5	—
CPS1	c.3+12249del	chr2:210489999	chr2:210489999	Heterozygous	Tier2	Not reviewed	—	—
CPSF3	p.Lys473Arg	chr2:9452935	rs1128215	Heterozygous	Tier2	Likely benign	20.2	0.065
CR1	p.Thr895Ala	chr1:207550815	rs147353507	Homozygous	Tier2	Not reviewed	6.1	0.063
CRACR2A	c.1602+6G>A	chr12:3638118	rs145536162	Heterozygous	Tier3	Likely benign	2.5	—
CSF3R	p.Glu808Lys	chr1:36466446	rs146617729	Heterozygous	Tier2	Benign/Likely benign	23.2	0.086
CSF3R	c.1865-65G>A	chr1:36467716	rs3917999	Heterozygous	Tier2	Benign	2.9	—
CTH	p.Thr67Ile	chr1:70415987	rs28941785	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	25.5	0.931
CTU2	p.Thr424Ile	chr16:88714656	rs373776054	Heterozygous	Tier2	Uncertain significance	10.4	0.128
CUBN	p.Pro1975Leu	chr10:16937594	rs41289303	Heterozygous	Tier3	Benign	25.3	0.341
CUX2	c.301+6815A>C	chr12:111270654	rs1265564	Heterozygous	Tier2	Not reviewed	0.7	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
CYP1A1	p.Thr461Asn	chr15:74720646	rs1799814	Heterozygous	Tier3	Benign	0.6	0.118
CYP21A2	p.Ser494Asn	chr6:32041127	rs6473	Heterozygous	Tier2	Benign	0.2	0.076
CYP2D6	p.Lys281del	chr22:42128174	rs5030656	Heterozygous	Tier3	Likely benign|other	—	—
CYP4A22	p.Leu72Ile	chr1:47140798	rs56258513	Heterozygous	Tier2	Not reviewed	15.4	0.135
DBH	p.Asn201Ser	chr9:133642322	rs45465204	Heterozygous	Tier3	Benign	10.7	0.184
DCAF17	c.*1841G>T	chr2:171482955	rs114519296	Heterozygous	Tier2	Conflicting classifications of pathogenicity	4.3	—
DCHS2	p.Glu3241Asp	chr4:154234929	rs753533849	Heterozygous	Tier2	Not reviewed	20.5	0.280
DCPS	p.Asp74Gly	chr11:126306589	rs145173937	Heterozygous	Tier2	Not reviewed	20.8	0.036
DDR2	c.-1310T>C	chr1:162630329	rs7544972	Heterozygous	Tier2	Not reviewed	0.4	—
DEFA5	p.Arg71His	chr8:7055504	rs7839771	Heterozygous	Tier3	Benign	0.0	0.049
DHFR	c.*1874C>T	chr5:80627213	rs918868210	Heterozygous	Tier2	Not reviewed	3.8	—
DMAP1	p.Glu239Gly	chr1:44218751	chr1:44218751	Heterozygous	**Tier1**	Not reviewed	29.7	0.377
DNAH3	c.5884+7486C>G	chr16:21012138	rs2087632048	Heterozygous	Tier2	Not reviewed	8.3	—
DOCK2	p.Ala1816Ser	chr5:170082811	rs144315682	Heterozygous	Tier2	Benign/Likely benign	0.3	0.016
DSC1	p.Gly740Arg	chr18:31132588	rs147041330	Heterozygous	Tier2	Not reviewed	25.8	0.287
DTNBP1	p.Pro272Ser	chr6:15523217	rs17470454	Heterozygous	Tier3	Benign/Likely benign	21.3	0.134
DYNLL2	c.*1936A>G	chr17:58091215	rs539751637	Heterozygous	Tier2	Not reviewed	18.6	—
EIF2AK1	p.Leu319His	chr7:6041055	rs34909691	Heterozygous	Tier3	Benign	7.4	0.028

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
ELOVL4	p.Glu272Gln	chr6:79916739	rs148919174	Heterozygous	Tier3	Benign/Likely benign	10.9	0.016
EMC1	c.96-708A>G	chr1:19245738	rs61764877	Homozygous	Tier2	Not reviewed	1.8	—
EML1	c.67+28559T>G	chr14:99822102	rs117232270	Heterozygous	Tier2	Not reviewed	4.4	—
EMSY	c.877-501T>A	chr11:76472063	rs368463439	Heterozygous	Tier2	Not reviewed	13.0	—
EPHA2	p.Arg721Gln	chr1:16132227	rs116506614	Heterozygous	Tier2	Conflicting classifications of pathogenicity	28.2	0.460
ERICH3	p.Asp1463His	chr1:74571323	rs12131196	Heterozygous	Tier2	Not reviewed	6.0	0.030
ETNK2	c.518+1064C>T	chr1:204148639	rs78068871	Heterozygous	Tier2	Not reviewed	1.3	—
EXOSC8	p.Pro87His	chr13:37005941	rs117135638	Heterozygous	Tier3	Benign	25.5	0.662
EYA2	c.-11+2233G>A	chr20:46897220	rs746126704	Heterozygous	Tier2	Not reviewed	18.4	—
F13A1	c.1908+5631T>C	chr6:6161827	rs6928884	Heterozygous	Tier2	Not reviewed	6.9	—
F5	p.Leu1397Phe	chr1:169540901	rs13306334	Heterozygous	Tier3	Benign/Likely benign	0.7	0.033
FAAH2	p.Arg42Trp	chrX:57286949	rs146874627	Heterozygous	Tier2	Benign	13.3	0.103
FAAP24	p.Ser126Phe	chr19:32974193	rs36017455	Heterozygous	**Tier1**	Benign	25.7	—
FAHD1	p.Ala102Ser	chr16:1827542	rs200357596	Heterozygous	**Tier1**	Not reviewed	27.2	0.715
FAM135A	p.Asp619Gly	chr6:70524940	rs1366288688	Heterozygous	Tier2	Not reviewed	3.7	0.073
FAM50A	p.Ala137Val	chrX:154446528	rs149558328	Heterozygous	Tier2	Not reviewed	0.1	0.007
FASN	p.Arg2125Trp	chr17:82081634	rs141141382	Heterozygous	Tier2	Likely benign	22.7	0.088
FAT1	p.Met1149Thr	chr4:186663433	rs77280820	Heterozygous	Tier2	Likely benign	20.7	0.104
FCGR1A	p.Asp324Asn	chr1:149791362	rs1050204	Heterozygous	Tier2	Not reviewed	7.1	0.049

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
FCGR3B	p.Ser36Arg	chr1:161629989	rs200688856	Heterozygous	Tier2	Not reviewed	0.6	0.016
FHIT	c.104-75688A>G	chr3:60089840	rs115894613	Heterozygous	Tier2	Not reviewed	1.7	—
FHIT	c.103+176862del	chr3:60359998	chr3:60359998	Heterozygous	Tier2	Not reviewed	—	—
FHIT	c.103+86860del	chr3:60450000	chr3:60450000	Heterozygous	Tier2	Not reviewed	—	—
FHIT	c.-17-10312G>C	chr3:60547291	rs114893898	Heterozygous	**Tier1**	Not reviewed	9.7	—
FLG	p.Glu2074Gln	chr1:152308666	rs141120727	Heterozygous	Tier3	Benign/Likely benign	0.0	0.033
FLG	p.His1880Asn	chr1:152309248	rs149817134	Heterozygous	Tier3	Benign/Likely benign	20.9	0.076
FLG	p.Thr581Ser	chr1:152313145	rs145627745	Heterozygous	Tier3	Benign/Likely benign	0.0	0.028
FLG2	p.Tyr881Ser	chr1:152355144	rs79239476	Heterozygous	Tier2	Not reviewed	0.0	0.016
FLG2	p.Ser877Gly	chr1:152355157	rs144938115	Heterozygous	Tier2	Not reviewed	0.0	0.012
FOXO1	p.Gly104Ala	chr13:40665902	rs963605898	Heterozygous	Tier2	Likely benign	12.6	0.282
FOXO3	c.621+9_621+18del	chr6:108569989	chr6:108569989	Heterozygous	Tier2	Not reviewed	—	—
FOXO3	c.621+10_621+18del	chr6:108569990	chr6:108569990	Heterozygous	Tier2	Not reviewed	—	—
FTCD	p.Glu128Lys	chr21:46151966	rs61729373	Heterozygous	Tier2	Benign	20.2	0.211
FZR1	p.Asn143=	chr19:3527021	rs138835338	Heterozygous	Tier2	Benign	0.1	—
GABRA1	c.*1381T>C	chr5:161898803	rs187522588	Heterozygous	Tier2	Conflicting classifications of pathogenicity	13.8	—
GABRB3	c.240+5408G>A	chr15:26766994	rs3212335	Heterozygous	Tier2	Not reviewed	5.9	—
GAN	c.1612+12A>G	chr16:81377340	rs150344737	Heterozygous	Tier2	Conflicting classifications of	0.5	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
						pathogenicity		
GATD3	p.Val6Ala	chr21:44133714	rs968714	Heterozygous	Tier2	Not reviewed	0.7	0.007
GATD3	p.Ser145Arg	chr21:44202056	rs6518351	Heterozygous	Tier2	Not reviewed	15.6	—
GDAP1	c.-18+8599A>G	chr8:74330187	rs2975212	Heterozygous	Tier2	Not reviewed	0.6	—
GDAP1	c.-18+12339A>G	chr8:74333927	rs2981271	Heterozygous	Tier2	Not reviewed	0.3	—
GDAP1	c.-18+14480A>G	chr8:74336068	rs2975206	Heterozygous	Tier2	Not reviewed	17.1	—
GFM1	p.Val664Ile	chr3:158690243	rs62288347	Heterozygous	Tier3	Benign/Likely benign	22.6	0.181
GJC2	p.Met147Leu	chr1:228158197	chr1:228158197	Heterozygous	Tier2	Not reviewed	22.8	0.445
GLRA3	p.Ala34Glu	chr4:174788914	rs748400564	Heterozygous	Tier2	Uncertain significance	21.9	0.180
GNAO1	c.*713del	chr16:56356775	chr16:56356775	Heterozygous	Tier2	Not reviewed	—	—
GNB2	c.-89-715T>C	chr7:100675462	rs117960080	Heterozygous	Tier2	Not reviewed	21.5	—
GNPAT	c.569-11del	chr1:231265279	rs199905093	Heterozygous	Tier2	Conflicting classifications of pathogenicity	—	—
GP5	p.Met289Ile	chr3:194397416	rs142440028	Heterozygous	Tier2	Not reviewed	17.3	0.027
GPAT2	p.Val555Ile	chr2:96024451	rs775199489	Homozygous	Tier2	Uncertain significance	10.1	0.096
GPER1	p.Ser373Arg	chr7:1092847	rs138657567	Heterozygous	Tier2	Not reviewed	0.7	0.091
GPNMB	c.224-16C>G	chr7:23254153	rs199346	Heterozygous	Tier2	Not reviewed	0.8	—
GRIN2A	c.1122+16340T>C	chr16:9874646	rs7204266	Heterozygous	Tier2	Not reviewed	4.2	—
GRIN3B	p.Pro736Ala	chr19:1007863	rs144432988	Heterozygous	Tier2	Uncertain significance	24.4	0.221

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
GUF1	p.Arg250Cys	chr4:44686523	rs116062572	Heterozygous	Tier2	Conflicting classifications of pathogenicity	22.1	0.035
HACD4	p.Leu91Phe	chr9:21016010	rs61736077	Heterozygous	Tier2	Not reviewed	24.9	0.219
HEPACAM	c.804-242G>C	chr11:124923060	rs11605766	Heterozygous	**Tier1**	Likely benign	14.2	—
HIVEP2	c.-640-6984A>G	chr6:142844031	chr6:142844031	Heterozygous	Tier2	Not reviewed	16.8	—
HLA-A	p.Pro217Ala	chr6:29944151	rs1059563	Heterozygous	Tier2	no classification for the single variant	15.0	0.095
HLA-A	p.Ile218Val	chr6:29944154	rs9260179	Heterozygous	Tier2	no classification for the single variant	0.0	0.016
HLA-A	c.*574G>A	chr6:29948263	rs9256984	Homozygous	Tier2	Uncertain significance	3.5	—
HLA-DQA1	c.-355-2A>G	chr6:32631940	rs9272089	Heterozygous	Tier2	Not reviewed	9.9	—
HLA-DQA1	p.Glu73Asp	chr6:32641446	rs9272697	Heterozygous	Tier2	Not reviewed	8.3	0.031
HLA-DQA1	p.Phe74Leu	chr6:32641447	rs9272698	Homozygous	Tier2	Not reviewed	13.2	0.097
HLA-DQB1	p.Thr103Asn	chr6:32664869	rs9274387	Heterozygous	Tier2	Not reviewed	0.0	0.158
HLA-DRB5	p.Val265Leu	chr6:32517747	rs77867566	Heterozygous	Tier2	Not reviewed	2.0	0.218
HLA-DRB5	p.Val265Leu	chr6:32517747	rs77867566	Heterozygous	Tier2	Not reviewed	2.0	0.212
HLA-DRB5	p.His260Gln	chr6:32518061	rs112835078	Homozygous	Tier2	Not reviewed	1.0	0.307
HLA-DRB5	p.Lys127Gln	chr6:32519643	rs138786370	Heterozygous	Tier2	Not reviewed	0.3	0.025
HLA-DRB5	c.371-392G>A	chr6:32520043	rs796105724	Heterozygous	**Tier1**	Not reviewed	8.4	—
HLA-DRB5	c.371-393T>G	chr6:32520044	rs796784269	Heterozygous	**Tier1**	Not reviewed	2.8	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
HLA-DRB5	p.His110Tyr	chr6:32521947	rs142908883	Heterozygous	Tier2	Not reviewed	0.3	0.093
HLA-DRB5	p.Ala103Gly	chr6:32521967	rs1059598	Heterozygous	Tier2	Not reviewed	0.1	0.109
HLA-DRB5	p.Ala102Gly	chr6:32521970	rs138927633	Heterozygous	Tier2	Not reviewed	15.1	0.118
HLA-DRB5	p.Asp99Glu	chr6:32521978	rs41559913	Homozygous	Tier2	Not reviewed	0.0	0.033
HLA-DRB5	p.Asp99Gly	chr6:32521979	rs41545413	Heterozygous	Tier2	Not reviewed	2.0	0.193
HLA-DRB5	p.Asp99His	chr6:32521980	rs41547217	Homozygous	Tier2	Not reviewed	0.2	0.177
HLA-DRB5	c.101-1G>A	chr6:32522175	rs201576424	Heterozygous	**Tier1**	Not reviewed	33.0	—
HLA-DRB5	c.100+17G>A	chr6:32530108	rs774541765	Heterozygous	**Tier1**	Not reviewed	6.7	—
HLA-DRB5	c.-123C>A	chr6:32530347	rs1188570510	Heterozygous	Tier2	Not reviewed	4.5	—
HLCS	c.1438-1318C>T	chr21:36931751	rs2409866	Heterozygous	Tier2	Not reviewed	1.1	—
HOXA13	p.Pro59Ala	chr7:27199903	rs747262113	Heterozygous	Tier2	Conflicting classifications of pathogenicity	19.3	0.307
HOXC10	p.Ser156Cys	chr12:53985726	rs149717368	Heterozygous	Tier2	Uncertain significance	24.7	0.527
HPSE	p.Phe236Cys	chr4:83310857	rs61755719	Heterozygous	Tier2	Likely benign	17.9	0.051
HSD17B1	p.Ser223Asn	chr17:42554533	rs146195480	Heterozygous	Tier2	Not reviewed	0.9	0.067
HSD17B6	p.Ile249Val	chr12:56787133	rs146253468	Heterozygous	Tier2	Uncertain significance	0.0	0.135
HSPA1B	p.Arg311Ter	chr6:31828881	rs1206932739	Heterozygous	Tier2	Not reviewed	36.0	—
HUWE1	p.Arg2992His	chrX:53551387	rs782317854	Heterozygous	Tier2	Uncertain significance	26.4	0.255
ICOSLG	p.Val128Ile	chr21:44236891	rs11558819	Heterozygous	Tier2	Benign/Likely benign	3.2	0.042

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
IFI27	p.Val26Ile	chr14:94111758	rs775606082	Heterozygous	Tier2	Not reviewed	2.5	0.015
IFIT3	p.Leu442Arg	chr10:89339980	rs116926108	Heterozygous	Tier2	Not reviewed	23.4	0.207
IFNA10	p.Val166Ile	chr9:21206602	rs111996205	Heterozygous	Tier2	Not reviewed	0.3	0.058
IFNA16	p.Thr124Ile	chr9:21216935	rs28368161	Heterozygous	Tier3	Benign	0.0	0.001
IFT122	p.Arg1043=	chr3:129514530	rs76881473	Heterozygous	Tier2	Conflicting classifications of pathogenicity	1.1	—
IGHV4-4	chr14:106012117	chr14:106012117	rs539822603	Heterozygous	Tier2	Not reviewed	33.0	—
IL17RC	p.Ala324Thr	chr3:9928397	rs115461448	Heterozygous	Tier3	Benign	23.2	0.091
IL17RC	c.1522+1G>C	chr3:9932859	rs148575246	Heterozygous	**Tier1**	Benign/Likely benign	33.0	—
IL1RAPL1	c.82+105560T>C	chrX:28894985	rs28826293	Heterozygous	Tier2	Not reviewed	2.2	—
IL1RAPL1	c.82+224347A>G	chrX:29013772	rs12843979	Homozygous	Tier2	Not reviewed	1.1	—
INO80D	p.Arg299Gln	chr2:206056266	rs144619437	Heterozygous	Tier2	Benign	21.7	0.048
INVS	c.273+37788A>G	chr9:100164337	rs863668	Homozygous	Tier2	Not reviewed	2.7	—
IQCB1	p.Glu481Lys	chr3:121772683	rs140630401	Heterozygous	Tier2	Conflicting classifications of pathogenicity	26.7	0.556
ITGA4	p.Val824Ala	chr2:181529581	rs1143675	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.4	0.104
ITGA4	c.*2362C>T	chr2:181537889	rs148226735	Heterozygous	Tier2	Uncertain significance	7.2	—
ITGB3BP	c.254+4779G>A	chr1:63473985	rs1190436510	Homozygous	Tier2	Not reviewed	7.2	—
ITGB8	c.-2181_-2180del	chr7:20329608	chr7:20329608	Heterozygous	Tier2	Not reviewed	—	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
ITPR2	p.Ser685=	chr12:26657844	rs2291264	Heterozygous	Tier3	Benign	0.3	—
KANSL2	p.Met370Ile	chr12:48660483	rs780890886	Heterozygous	Tier2	Uncertain significance	22.4	0.165
KCNB1	p.Ser857Asn	chr20:49372990	rs34280195	Heterozygous	Tier3	Benign	23.2	0.381
KCNB1	p.Thr616Ser	chr20:49373713	rs2229006	Heterozygous	Tier3	Benign	15.7	0.110
KCND3	c.1107-86635A>G	chr1:111873741	rs2788612	Homozygous	Tier2	Not reviewed	18.2	—
KCNK13	p.Gly331Arg	chr14:90184767	rs116117726	Heterozygous	Tier2	Not reviewed	19.9	0.085
KCNMB2	c.-68+29835C>T	chr3:178302841	rs1357843772	Heterozygous	Tier2	Not reviewed	6.9	—
KCNU1	p.Gln1119Arg	chr8:36935826	rs756494970	Heterozygous	Tier2	Uncertain significance	2.3	0.009
KDM6B	p.Val209Leu	chr17:7846654	rs148641957	Heterozygous	Tier2	Conflicting classifications of pathogenicity	18.4	0.029
KIAA1191	c.-276A>G	chr5:176361710	rs1768029962	Heterozygous	Tier2	Not reviewed	20.7	—
LAMA5	p.Pro3536Ser	chr20:62310306	rs41307203	Heterozygous	Tier3	Benign	8.0	0.147
LAMA5	p.Ala3197=	chr20:62311964	rs114698761	Heterozygous	Tier2	Benign	0.2	—
LAMA5	p.Arg3165Gln	chr20:62312183	rs140181393	Heterozygous	Tier3	Benign	12.4	0.113
LAMA5	p.Arg2123His	chr20:62322147	rs374836875	Heterozygous	Tier2	Benign/Likely benign	16.6	0.026
LAMA5	p.Arg1903Lys	chr20:62324140	rs116895933	Heterozygous	Tier3	Benign/Likely benign	1.0	0.054
LAMA5	p.Pro1260Ala	chr20:62330817	rs780286336	Heterozygous	Tier2	Uncertain significance	15.3	0.108
LAMB2	p.Ala1765Thr	chr3:49121330	rs74951356	Heterozygous	Tier3	Benign/Likely benign	13.9	0.046

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
LAMB2	p.Arg1436His	chr3:49122970	rs377526198	Heterozygous	Tier2	Uncertain significance	21.3	0.053
LAMB2	p.Leu769=	chr3:49126004	rs147076626	Heterozygous	Tier2	Conflicting classifications of pathogenicity	5.5	—
LCA5	c.-398G>C	chr6:79537371	rs370115829	Heterozygous	Tier2	Conflicting classifications of pathogenicity	7.2	—
LDAH	p.Arg151Gln	chr2:20774826	rs114010802	Heterozygous	Tier2	Benign	0.9	0.047
LDHAL6A	c.592+1G>A	chr11:18475640	rs145833201	Heterozygous	**Tier1**	Not reviewed	31.0	—
LEP	c.*653A>C	chr7:128255416	rs201341056	Heterozygous	Tier2	Uncertain significance	0.8	—
LIFR	p.Arg780Cys	chr5:38485978	rs553912841	Heterozygous	Tier2	Uncertain significance	28.6	0.166
LIFR	p.Tyr762His	chr5:38489129	rs200270315	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.4	0.240
LIFR	p.Ser723fs	chr5:38489995	chr5:38489995	Heterozygous	Tier2	Not reviewed	—	—
LIFR	p.Ser723fs	chr5:38489999	chr5:38489999	Heterozygous	Tier2	Not reviewed	—	—
LILRA6	p.Gln350Ter	chr19:54240484	rs2361802	Heterozygous	Tier2	Not reviewed	22.2	—
LILRA6	p.Arg205Trp	chr19:54241621	rs1052973	Heterozygous	Tier2	Not reviewed	16.3	0.015
LILRA6	p.Ser122Asn	chr19:54241869	rs1132595	Homozygous	Tier2	Not reviewed	0.0	—
LILRA6	p.Ala120Val	chr19:54241875	rs1132593	Homozygous	Tier2	Not reviewed	0.0	—
LILRA6	p.Ala120Ser	chr19:54241876	rs1132591	Homozygous	Tier2	Not reviewed	0.0	—
LILRA6	p.Met117Thr	chr19:54242031	rs62133127	Heterozygous	Tier2	Not reviewed	12.4	0.099

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
LILRA6	p.His20Arg	chr19:54242530	rs201396172	Homozygous	Tier2	Not reviewed	0.0	—
LILRB3	p.Tyr350Ter	chr19:54220736	rs16985497	Homozygous	Tier2	Not reviewed	23.2	—
LILRB3	p.Tyr350His	chr19:54220738	rs1052987	Homozygous	Tier2	Not reviewed	0.0	—
LILRB3	p.Gly349Trp	chr19:54220741	rs1052985	Homozygous	Tier2	Not reviewed	0.0	—
LILRB3	p.Arg348Trp	chr19:54220744	rs140963490	Homozygous	Tier2	Not reviewed	0.0	—
LILRB3	p.Thr318Ala	chr19:54221086	rs563416061	Homozygous	Tier2	Not reviewed	0.0	—
LILRB3	p.Ile317Met	chr19:54221087	rs529098038	Homozygous	Tier2	Not reviewed	8.8	—
LITAF	c.-5-858C>A	chr16:11557593	rs9923255	Heterozygous	Tier2	Not reviewed	1.5	—
LNPK	c.358-1181T>C	chr2:175965770	chr2:175965770	Heterozygous	Tier2	Not reviewed	8.1	—
LPA	p.Tyr2023Cys	chr6:160531784	rs41267807	Heterozygous	Tier3	Not reviewed	25.9	0.614
LRRK2	p.Pro1542Ser	chr12:40314059	rs33958906	Heterozygous	Tier3	Benign/Likely benign	24.7	0.228
LTF	c.1357+398C>T	chr3:46446042	rs766222508	Heterozygous	Tier2	Not reviewed	5.0	—
MACF1	c.21959-173C>T	chr1:39479625	rs2275767	Heterozygous	Tier2	Benign	3.4	—
MACROD1	c.665-35G>C	chr11:63999798	rs202065074	Heterozygous	Tier2	Not reviewed	5.8	—
MAGI2	c.1046-308C>T	chr7:78369521	rs7783086	Heterozygous	Tier2	Benign	0.3	—
MAK	c.-254G>C	chr6:10838527	rs72821595	Heterozygous	Tier2	Uncertain significance	6.3	—
MAPT	p.Gln305Arg	chr17:45983493	rs63750072	Heterozygous	Tier3	Benign/Likely benign	22.3	0.079
MC1R	p.Asp294His	chr16:89920138	rs1805009	Heterozygous	Tier2	Conflicting classifications of pathogenicity|risk factor	26.9	0.608

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
MCPH1	c.-1527_-1526del	chr8:6405127	chr8:6405127	Heterozygous	Tier2	Not reviewed	—	—
MDM4	p.Lys374Gln	chr1:204549329	rs41299595	Heterozygous	Tier2	Conflicting classifications of pathogenicity	21.4	0.047
MEF2C	c.638-1909G>A	chr5:88733810	rs412458	Heterozygous	Tier2	Not reviewed	0.3	—
MEF2C	c.-528_-526del	chr5:88883338	chr5:88883338	Heterozygous	Tier2	Conflicting classifications of pathogenicity	—	—
METTL9	c.-251C>G	chr16:21599483	rs1478461	Homozygous	Tier2	Not reviewed	7.0	—
MFSD12	p.Arg472Gln	chr19:3544814	rs200371465	Heterozygous	Tier2	Not reviewed	22.4	0.101
MGST3	p.Trp139Ser	chr1:165655461	rs199915214	Heterozygous	Tier2	Not reviewed	29.2	0.496
MID1IP1	p.Gly182Asp	chrX:38805491	rs143008624	Heterozygous	**Tier1**	Benign	26.1	0.630
MMADHC	p.Lys29Asn	chr2:149582194	rs61750442	Heterozygous	Tier2	Benign/Likely benign	23.3	0.327
MMP19	p.Arg456Gln	chr12:55837196	rs150724096	Heterozygous	Tier2	Likely benign	21.5	0.097
MPDZ	p.Arg1229Gln	chr9:13147603	rs61753783	Heterozygous	Tier2	Uncertain significance	28.6	0.384
MPL	c.1654-10T>A	chr1:43352508	rs200460456	Heterozygous	Tier2	Conflicting classifications of pathogenicity	10.9	—
MPV17	c.*413A>C	chr2:27309499	rs575276701	Heterozygous	Tier2	Uncertain significance	4.5	—
MRPL21	c.396+31G>A	chr11:68896484	rs76789024	Heterozygous	Tier2	Not reviewed	6.4	—
MRPL23	p.Ala122Thr	chr11:1956322	rs12812	Heterozygous	Tier2	Not reviewed	0.0	0.008
MRPL23	p.Arg162Ser	chr11:1972745	rs190063	Heterozygous	Tier2	Not reviewed	8.7	0.011

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
MRPL34	p.Ala74Val	chr19:17306321	rs201529220	Heterozygous	Tier2	Not reviewed	20.7	0.033
MSLN	p.Arg557His	chr16:768452	rs150425699	Heterozygous	Tier3	Benign	16.4	0.034
MSRA	c.211+6549G>A	chr8:10214450	rs955247088	Heterozygous	Tier2	Not reviewed	16.8	—
MTHFD1	p.Lys134Arg	chr14:64415662	rs1950902	Heterozygous	Tier3	Benign	20.0	0.237
MTMR3	c.*63G>T	chr22:30025864	rs746220322	Heterozygous	Tier2	Not reviewed	16.1	—
MTR	p.Asp314Asn	chr1:236826841	rs2229274	Heterozygous	Tier3	Benign	22.5	0.120
MTR	c.*4347A>T	chr1:236901991	rs767109806	Heterozygous	Tier2	Uncertain significance	1.2	—
MUC16	p.Arg12256Gln	chr19:8916887	rs139868374	Heterozygous	Tier2	Benign	14.7	0.059
MX1	p.Arg265Gln	chr21:41441779	rs145851733	Heterozygous	Tier2	Uncertain significance	14.0	0.123
MXRA8	p.Asp396Asn	chr1:1354066	rs150058708	Heterozygous	Tier2	Likely benign	23.3	0.101
MYO6	c.*1324A>G	chr6:75916336	rs191987417	Heterozygous	Tier2	Uncertain significance	10.5	—
MYOF	c.4719+179G>A	chr10:93333579	rs12782443	Heterozygous	Tier2	Not reviewed	12.6	—
MYOM1	c.1174+2180C>T	chr18:3171758	rs12965146	Heterozygous	Tier2	Not reviewed	3.4	—
NAIP	p.Val535Met	chr5:70985446	rs1423904967	Homozygous	Tier2	Not reviewed	5.1	0.035
NFIA	c.96+3376C>G	chr1:60868842	rs1375339493	Heterozygous	Tier2	Not reviewed	15.7	—
NFIA	c.626-26539G>A	chr1:61305973	rs2806434	Heterozygous	Tier2	Not reviewed	1.8	—
NIPSNAP2	c.712+1G>T	chr7:55994989	rs748801121	Heterozygous	**Tier1**	Not reviewed	33.0	—
NLRC4	p.Ala883Thr	chr2:32235536	rs780229500	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.4	0.018

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NLRC4	p.Tyr594Asp	chr2:32250084	chr2:32250084	Heterozygous	Tier2	Not reviewed	22.4	0.261
NLRP2	p.Thr516Ala	chr19:54983244	rs61735082	Heterozygous	Tier3	Likely benign	0.0	0.070
NLRP2	p.Glu522Gly	chr19:54983263	rs61735083	Heterozygous	Tier3	Likely benign	8.7	0.097
NLRP4	p.Gly638Arg	chr19:55861441	rs143589259	Heterozygous	Tier2	Not reviewed	1.9	0.201
NLRP9	p.Arg899Cys	chr19:55711948	rs148511077	Heterozygous	Tier3	Benign	1.7	0.128
NME2	c.-5+1G>A	chr17:51166498	rs115848216	Heterozygous	Tier2	Not reviewed	16.9	—
NPC2	c.441+1G>A	chr14:74480701	rs140130028	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	23.2	0.207
NPHS1	c.-57G>T	chr19:35851894	rs140404523	Heterozygous	Tier2	Conflicting classifications of pathogenicity	5.7	—
NPY4R	p.Cys240Arg	chr10:46461918	rs1352903918	Heterozygous	Tier3	Benign	16.7	—
NQO1	p.Pro187Ser	chr16:69711242	rs1800566	Heterozygous	Tier2	Uncertain significance	24.2	0.366
NR4A1	p.Thr143Met	chr12:52054756	rs373652442	Heterozygous	Tier2	Uncertain significance	24.8	0.528
NRP1	p.Asp636Asn	chr10:33197668	rs145594886	Heterozygous	Tier2	Uncertain significance	22.9	0.392
NRXN1	c.4128+39765G>A	chr2:50013506	rs3796036	Homozygous	Tier2	Not reviewed	0.1	—
NRXN1	c.3365-51910A>G	chr2:50288880	rs1356888	Heterozygous	Tier2	Not reviewed	1.8	—
NRXN1	c.-1273G>A	chr2:51032332	rs186103615	Heterozygous	Tier2	Conflicting classifications of pathogenicity	14.1	—
NRXN3	c.2275+65009A>T	chr14:78875353	rs866714403	Heterozygous	Tier2	Not reviewed	15.4	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL
NTNG1	c.1391-11644G>A	chr1:107468967	rs11185120	Heterozygous	Tier2	Not reviewed	0.8	—
OGFOD3	c.380+1G>A	chr17:82411454	rs150938774	Heterozygous	**Tier1**	Not reviewed	34.0	—
ONECUT1	c.1105+3262C>T	chr15:52785518	rs34939760	Heterozygous	Tier2	Not reviewed	1.5	—
OPA3	c.*1070C>T	chr19:45552444	rs146989248	Heterozygous	Tier2	Uncertain significance	1.6	—
ORAI1	p.Ser218Gly	chr12:121641389	rs3741596	Heterozygous	Tier3	Benign	1.1	—
ORC1	p.Ala372Val	chr1:52389289	rs3087476	Heterozygous	Tier3	Benign	12.2	0.122
P3H2	c.824-1035T>A	chr3:189990073	rs55848820	Homozygous	Tier2	Not reviewed	9.0	—
P3H3	p.Arg199Ter	chr12:6829955	rs1225283465	Heterozygous	Tier2	Not reviewed	35.0	—
P4HA2	c.-19+2739_-19+2740dup	chr5:132225051	chr5:132225051	Heterozygous	Tier2	Not reviewed	—	—
PACS2	p.Pro725Leu	chr14:105391685	rs147461490	Heterozygous	Tier3	Benign	23.6	0.157
PAK4	p.Arg153Gln	chr19:39173171	rs754471124	Heterozygous	Tier2	Uncertain significance	22.3	0.173
PALLD	c.1965-12907G>A	chr4:168878015	rs780516159	Heterozygous	Tier2	Conflicting classifications of pathogenicity	10.2	0.062
PAPSS1	c.1736+4598T>A	chr4:107627033	chr4:107627033	Heterozygous	Tier2	Not reviewed	18.6	—
PARPBP	p.Asn81Asp	chr12:102148317	chr12:102148317	Heterozygous	Tier2	Not reviewed	15.6	0.016
PCDH15	c.4368-1931C>T	chr10:53822161	rs111033445	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.5	—
PCDHA9	p.Gln534Glu	chr5:140850095	rs139719626	Heterozygous	Tier2	Not reviewed	15.4	0.119
PCDHGA5	p.Ser184Arg	chr5:141364882	rs199576947	Heterozygous	Tier2	Not reviewed	16.4	0.064
PGK1	p.Lys133Glu	chrX:78114140	chrX:78114140	Heterozygous	**Tier1**	Not reviewed	22.7	0.543

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PHIP	c.3379+491A>G	chr6:78965212	rs10943608	Heterozygous	Tier2	Not reviewed	1.3	—
PHKB	p.Met185Ile	chr16:47515562	rs56257827	Heterozygous	Tier3	Benign/Likely benign	26.2	0.773
PI4KA	p.Ile339Val	chr22:20811023	rs61752248	Heterozygous	Tier3	Likely benign	23.6	0.016
PI4KA	c.530-124G>T	chr22:20820024	rs165691	Heterozygous	Tier3	Benign	0.4	—
PIEZO1	c.65-6410A>G	chr16:88755889	rs3743975	Homozygous	Tier2	Not reviewed	1.8	—
PIKFYVE	c.-10+5G>A	chr2:208266420	rs541362896	Heterozygous	Tier3	Likely benign	18.1	—
PKD2	c.*1010G>A	chr4:88076704	rs144024108	Heterozygous	Tier2	Uncertain significance	0.1	—
PKHD1	c.6122-5904A>G	chr6:51918480	rs12199884	Heterozygous	Tier2	Not reviewed	1.3	—
PKP4	c.-5-16141T>C	chr2:158517039	rs1436595521	Heterozygous	Tier2	Not reviewed	20.1	—
PLA2G4E	p.Val716Met	chr15:41984589	rs774426285	Heterozygous	Tier2	Uncertain significance	1.7	0.017
PLA2G4F	c.601+417C>T	chr15:42152271	rs116885045	Heterozygous	Tier2	Not reviewed	0.0	—
PLB1	c.-114+4401G>A	chr2:28461598	rs56116810	Heterozygous	**Tier1**	Not reviewed	3.5	—
PLCD1	p.Asp231Asn	chr3:38011313	rs61755441	Heterozygous	Tier3	Benign	19.9	0.181
PLEKHG2	p.Pro453Arg	chr19:39420811	rs34975345	Homozygous	Tier3	Benign	22.5	0.083
PLEKHG2	p.Ala770Thr	chr19:39423362	rs111487768	Heterozygous	Tier2	Benign/Likely benign	25.0	0.378
PLK4	p.Gly6Glu	chr4:127881151	rs149003893	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	28.5	0.356
PMS2	p.Ala520Val	chr7:5987206	rs63751300	Heterozygous	Tier2	Conflicting classifications of pathogenicity	1.4	0.090

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PNKP	c.1127-8C>T	chr19:49862113	rs3739203	Heterozygous	Tier3	Benign/Likely benign	5.3	—
PNKP	p.Pro20Ser	chr19:49867147	rs3739168	Heterozygous	Tier2	Benign/Likely benign	17.9	0.035
POLDIP2	p.Ala36Val	chr17:28357342	rs11555978	Heterozygous	Tier3	Likely benign	15.0	—
PPM1D	p.Ser376Phe	chr17:60656708	rs61739281	Heterozygous	Tier2	Uncertain significance	24.0	0.194
PRICKLE1	c.-48-11679G>C	chr12:42484243	rs538816869	Heterozygous	Tier2	Not reviewed	9.5	—
PRKD1	c.33+60419_33+60420del	chr14:29987294	chr14:29987294	Heterozygous	Tier2	Not reviewed	—	—
PRKD1	c.33+31108A>G	chr14:30016602	rs1360253541	Heterozygous	Tier2	Not reviewed	19.2	—
PRKDC	c.-766G>A	chr8:47960892	rs17334305	Heterozygous	Tier2	Uncertain significance	2.4	—
PRKN	c.8-14826C>T	chr6:162458299	rs771567497	Heterozygous	Tier2	Not reviewed	2.3	—
PRODH	c.1251+419G>A	chr22:18919032	rs449138	Homozygous	Tier2	Not reviewed	0.5	—
PRODH	p.Trp185Arg	chr22:18925165	rs4819756	Heterozygous	Tier2	Benign	7.6	0.124
PROS1	c.*686G>A	chr3:93873559	rs550909963	Heterozygous	Tier2	Uncertain significance	0.9	—
PRR12	p.Ala649Val	chr19:49596281	rs371012654	Heterozygous	Tier2	Likely benign	22.3	0.202
PTPRC	p.Asp123His	chr1:198699632	rs41269905	Heterozygous	Tier3	Benign/Likely benign	22.8	0.070
PTPRC	p.Ile296Leu	chr1:198706934	rs2230606	Heterozygous	Tier3	Benign/Likely benign	0.0	0.043
PTPRC	p.His570Gln	chr1:198722466	rs12136658	Heterozygous	Tier3	Benign/Likely benign	0.0	0.055
PUS3	p.Pro179Ser	chr11:125895633	rs35681508	Heterozygous	Tier3	Benign	24.5	0.262

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PXDNL	p.Cys1258Ter	chr8:51372000	rs117752382	Heterozygous	Tier2	Benign	36.0	—
PYROXD2	c.127+23G>C	chr10:98414986	rs1475191215	Heterozygous	Tier2	Not reviewed	9.4	—
RAD1	p.Thr104Ser	chr5:34911809	rs1805328	Heterozygous	Tier3	Benign	14.9	0.055
RAD51B	c.756+102812del	chr14:67989999	chr14:67989999	Heterozygous	Tier2	Not reviewed	—	—
RANBP2	p.Gly2692Asp	chr2:108772543	rs111958052	Heterozygous	Tier2	Likely benign	13.6	0.065
RAP1GDS1	p.Val105Met	chr4:98352553	rs61758810	Heterozygous	Tier2	Likely benign	20.3	0.320
RCC1L	p.Arg15Trp	chr7:75073695	rs201728682	Heterozygous	Tier3	Benign	22.9	—
REN	c.-211+1G>A	chr1:204190231	rs76834242	Heterozygous	Tier2	Benign	17.4	—
RFXAP	p.Met137Thr	chr13:36819767	rs193240312	Heterozygous	Tier3	Benign/Likely benign	14.2	0.027
RHCE	p.Pro103Ser	chr1:25408711	rs676785	Homozygous	Tier2	not provided	0.0	0.015
RHCE	p.Asn68Ser	chr1:25408815	rs1053344	Homozygous	Tier2	Not reviewed	3.7	0.043
RHCE	p.Leu60Ile	chr1:25408840	rs181860403	Homozygous	Tier2	not provided	12.0	0.057
RNF152	p.Lys108Asn	chr18:61816140	rs369621346	Heterozygous	Tier2	Uncertain significance	18.8	0.261
ROBO1	c.500-57835_500-57833del	chr3:78804734	chr3:78804734	Heterozygous	Tier2	Not reviewed	—	—
RTN3	p.Asp501His	chr11:63720003	rs7936660	Heterozygous	Tier2	Not reviewed	25.6	0.236
RUNX1	c.59-63960A>G	chr21:34956923	rs148127564	Heterozygous	Tier2	Not reviewed	10.7	—
SAA1	c.91+760T>C	chr11:18267738	rs111755795	Heterozygous	Tier2	Not reviewed	3.7	—
SACS	p.Ile2575=	chr13:23336151	rs200735789	Heterozygous	Tier2	Conflicting classifications of pathogenicity	9.0	—
SCAF4	p.Phe341Leu	chr21:31696160	rs117061499	Heterozygous	Tier2	Likely benign	24.1	0.096

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SCN3A	c.-248+3133G>T	chr2:165200690	rs116793108	Heterozygous	Tier2	Not reviewed	20.0	—
SCNN1A	p.Thr42=	chr12:6374658	rs138995556	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.9	—
SEC23B	p.Val426Ile	chr20:18532706	rs41309927	Heterozygous	Tier3	Benign/Likely benign	18.2	0.122
SEPTIN9	c.76+23238A>C	chr17:77330435	rs892964	Heterozygous	Tier2	Not reviewed	2.3	—
SERPINA1	c.*80C>T	chr14:94378369	rs1243163	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.1	—
SERPINC1	p.Gln337=	chr1:173909694	rs5878	Heterozygous	Tier2	Benign	9.2	—
SERPINH1	c.*228A>G	chr11:75572311	rs189882479	Heterozygous	Tier2	Uncertain significance	3.5	—
SETD1B	p.Ala1488Thr	chr12:121823041	rs144871159	Heterozygous	Tier3	Benign	10.6	0.026
SETD5	c.-470_-468del	chr3:9397655	chr3:9397655	Homozygous	Tier2	Not reviewed	—	—
SFXN5	c.625+7137C>T	chr2:72981121	rs1363816061	Heterozygous	Tier2	Not reviewed	4.2	—
SIL1	c.-10-12945G>A	chr5:139140798	rs1187156016	Heterozygous	Tier2	Not reviewed	17.6	—
SIM2	p.Pro649Ala	chr21:36748033	rs1469534783	Heterozygous	Tier2	Not reviewed	2.1	0.075
SIMC1	c.130-22537G>A	chr5:176267117	rs2560173	Homozygous	Tier2	Not reviewed	3.2	—
SIRPA	p.Asp95Glu	chr20:1915304	rs138283486	Homozygous	Tier2	Not reviewed	0.8	0.299
SIRPA	p.Leu96Phe	chr20:1915305	rs149634649	Homozygous	Tier2	Not reviewed	0.0	0.072
SIRPA	p.Leu96Pro	chr20:1915306	rs146163282	Homozygous	Tier2	Not reviewed	0.2	0.033
SIRT2	p.Arg153His	chr19:38889130	rs144373891	Heterozygous	Tier2	Benign	26.0	0.231

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SLC12A5	p.Arg86Gly	chr20:46022977	rs58366483	Homozygous	Tier2	Benign/Likely benign	14.4	—
SLC12A6	c.-73+250A>G	chr15:34337082	rs141533243	Heterozygous	Tier3	Likely benign	15.8	—
SLC25A15	c.*1721C>T	chr13:40811388	rs536553279	Heterozygous	Tier2	Uncertain significance	6.6	—
SLC27A3	p.Leu173Pro	chr1:153776015	rs756978807	Heterozygous	**Tier1**	Not reviewed	27.2	0.411
SLC30A7	c.*3807T>C	chr1:100978664	chr1:100978664	Heterozygous	Tier2	Not reviewed	15.9	—
SLC4A2	p.Arg304Trp	chr7:151066937	rs371485247	Heterozygous	Tier2	Uncertain significance	26.5	0.172
SLC5A4	c.886-5G>A	chr22:32233039	rs9621415	Heterozygous	**Tier1**	Not reviewed	14.0	—
SLC6A4	c.1818+1348A>G	chr17:30201824	rs11080121	Heterozygous	Tier2	Not reviewed	5.6	—
SLC8A1	c.1808+48_1808+49del	chr2:40428426	chr2:40428426	Heterozygous	Tier2	Not reviewed	—	—
SLCO3A1	p.Ile560Val	chr15:92147149	rs141187463	Heterozygous	Tier2	Likely benign	23.0	0.180
SLF2	c.2513-423A>G	chr10:100938172	rs531379316	Heterozygous	Tier2	Uncertain significance	0.3	—
SLIT2	p.Gly909Asp	chr4:20567262	rs1020555095	Heterozygous	**Tier1**	Not reviewed	32.0	0.743
SMG1	p.Ala1209Ser	chr16:18863720	rs137863609	Heterozygous	Tier3	Benign	16.1	0.057
SMPD4	p.Pro217Leu	chr2:130172358	rs144353633	Heterozygous	Tier2	Not reviewed	27.6	0.303
SOX5	c.271-14760del	chr12:23860954	chr12:23860954	Heterozygous	Tier2	Not reviewed	—	—
SP110	p.Glu538Lys	chr2:230172938	rs149682257	Heterozygous	Tier2	Uncertain significance	13.9	0.267
SP110	c.1349-1241T>C	chr2:230179496	rs967006	Homozygous	Tier2	Not reviewed	3.4	—
SPAG16	c.1720+38410G>A	chr2:214187676	chr2:214187676	Heterozygous	Tier2	Not reviewed	0.7	—

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SPG11	p.Pro2345Leu	chr15:44564664	rs748206574	Heterozygous	Tier2	Uncertain significance	32.0	0.498
SPIDR	p.Met316Lys	chr8:47440392	rs201125764	Heterozygous	Tier2	Likely benign	19.8	0.131
SPRTN	p.Thr439Ser	chr1:231353206	rs62617126	Heterozygous	Tier2	Benign	0.1	0.003
SRD5A2	p.Ala49Thr	chr2:31580756	rs9282858	Heterozygous	Tier3	Benign	20.9	—
SSTR4	p.Ala245Asp	chr20:23036217	rs35136964	Heterozygous	Tier2	Not reviewed	24.0	0.419
ST3GAL3	p.Trp70fs	chr1:43829996	rs71579312	Homozygous	Tier2	Not reviewed	—	—
ST3GAL5	c.-91G>A	chr2:85888996	rs879798501	Heterozygous	Tier2	Uncertain significance	12.1	—
STAP1	c.730-41G>C	chr4:67593219	rs3816821	Heterozygous	Tier3	Benign	0.3	—
STOX1	p.Asn825Ile	chr10:68886270	rs41278532	Heterozygous	Tier2	Not reviewed	15.5	0.214
SUCLA2	c.*214+8995G>A	chr13:47896334	rs7334463	Homozygous	Tier2	Not reviewed	0.0	—
SUCLA2	c.*27-16018C>T	chr13:47923153	rs9591123	Homozygous	Tier2	Not reviewed	1.2	—
SUMF1	p.Asn352Lys	chr3:4362213	rs376126385	Heterozygous	**Tier1**	Uncertain significance	23.9	0.641
SYNPR	c.84+40453T>G	chr3:63319195	rs73123134	Heterozygous	**Tier1**	Not reviewed	6.8	—
SYT1	c.*833_*834dup	chr12:79449926	chr12:79449926	Heterozygous	Tier2	Not reviewed	—	—
SYT9	c.146-23864A>G	chr11:7279175	chr11:7279175	Heterozygous	Tier2	Not reviewed	16.5	—
TAC4	c.106-338G>A	chr17:49844495	chr17:49844495	Heterozygous	Tier2	Not reviewed	1.7	—
TANC2	c.-24+5272A>C	chr17:62972021	rs763655159	Heterozygous	Tier2	Not reviewed	14.2	—
TAP1	p.Met739Val	chr6:32845611	rs780766118	Heterozygous	Tier2	Uncertain significance	22.8	0.440

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TARS2	p.His345Arg	chr1:150497543	rs115390773	Heterozygous	Tier2	Benign/Likely benign	1.6	0.067
TAT	c.*409G>A	chr16:71567735	rs572677025	Heterozygous	Tier2	Uncertain significance	0.8	—
TBP	p.Gln95del	chr6:170561959	chr6:170561959	Heterozygous	Tier3	Benign	—	—
TCF20	p.Arg1438Cys	chr22:42210994	rs771336001	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.4	0.166
TCF7L2	p.Pro500Thr	chr10:113165610	rs77961654	Heterozygous	Tier3	Benign	22.6	0.378
TECPR2	p.Lys470dup	chr14:102432106	chr14:102432106	Heterozygous	Tier3	Likely benign	—	—
TECPR2	p.Ser1092Leu	chr14:102449828	rs72700618	Heterozygous	Tier2	Benign/Likely benign	27.6	0.407
TEK	p.Asn438Ile	chr9:27185615	rs189543659	Heterozygous	**Tier1**	Not reviewed	24.3	0.208
TEK	p.Val600Leu	chr9:27197488	rs35030851	Heterozygous	Tier3	Benign	17.9	0.058
THSD1	p.Lys768Arg	chr13:52377667	rs9536041	Heterozygous	Tier3	Benign	14.7	0.092
TICAM1	p.Pro367del	chr19:4817277	chr19:4817277	Heterozygous	Tier3	Likely benign	—	—
TIRAP	p.Ser180Leu	chr11:126292948	rs8177374	Heterozygous	Tier3	Benign	13.5	0.092
TLK2	c.1550+842G>A	chr17:62597516	rs2082566857	Heterozygous	Tier2	Not reviewed	11.1	—
TLN1	p.Gln1524Lys	chr9:35707793	chr9:35707793	Heterozygous	Tier2	Not reviewed	24.8	0.325
TLR1	p.His305Leu	chr4:38797918	rs3923647	Heterozygous	Tier3	Benign	17.6	0.079
TLR2	p.Phe217Ser	chr4:153703557	rs139227237	Heterozygous	Tier2	Benign/Likely benign	20.2	0.526
TMEM79	p.Arg127Cys	chr1:156285605	rs138888317	Heterozygous	Tier2	Not reviewed	25.5	0.235
TNC	p.Arg1698Gln	chr9:115046442	rs61734387	Heterozygous	Tier3	Benign	25.6	0.185

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TNC	p.Arg1066Cys	chr9:115073621	rs139280264	Heterozygous	Tier2	Likely benign	23.0	0.197
TNR	c.499+596_499+597del	chr1:175405623	chr1:175405623	Heterozygous	Tier2	Not reviewed	—	—
TOM1	c.1284+86A>G	chr22:35345870	rs133399	Heterozygous	Tier3	Benign	2.7	—
TP53BP1	p.Glu742Asp	chr15:43456382	rs150423877	Heterozygous	Tier2	Not reviewed	21.9	0.080
TP53RK	p.Arg123Gln	chr20:46687147	rs34983477	Heterozygous	Tier3	Benign	22.8	0.126
TPK1	c.614-47135A>C	chr7:144500798	rs1308701958	Heterozygous	Tier2	Not reviewed	12.9	—
TPM4	c.133-1860T>G	chr19:16080053	rs16981095	Heterozygous	Tier2	Not reviewed	3.0	—
TPTE2	p.Ser434Trp	chr13:19430469	rs149205749	Heterozygous	Tier2	Not reviewed	0.5	0.199
TRAV1-2	chr14:21641263	chr14:21641263	chr14:21641263	Heterozygous	Tier2	Not reviewed	—	—
TRIO	c.2046+9360A>G	chr5:14346087	rs62344980	Heterozygous	Tier2	Not reviewed	4.3	—
TRPC7	c.-438_-433dup	chr5:136365689	chr5:136365689	Heterozygous	Tier2	Not reviewed	—	—
TRPM1	p.Glu1303Lys	chr15:31002793	rs117855013	Heterozygous	Tier2	Benign/Likely benign	26.2	0.194
TST	p.Ser241Leu	chr22:37011199	rs140912272	Heterozygous	Tier2	Uncertain significance	15.8	0.178
TTC21B	p.Leu473Phe	chr2:165924648	rs2163649	Heterozygous	Tier3	Benign	14.1	0.111
UBE2A	c.-157-36745C>T	chrX:119536898	rs1967298	Heterozygous	Tier2	Not reviewed	1.5	—
UBR5	c.886+9G>A	chr8:102345387	rs760900237	Heterozygous	Tier2	Not reviewed	21.0	—
UCK2	c.99+808G>T	chr1:165828740	rs562989479	Heterozygous	**Tier1**	Not reviewed	6.9	—
UGT2B4	c.721+2T>C	chr4:69495139	rs377159596	Heterozygous	Tier2	Not reviewed	23.8	—
ULBP2	p.Ile237Leu	chr6:149947397	rs146872141	Heterozygous	Tier3	Benign	0.2	0.026
UQCC2	c.138+87G>T	chr6:33711462	rs7759182	Heterozygous	Tier2	Benign	15.7	—

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UQCRQ	c.*236C>T	chr5:132867818	rs116229809	Heterozygous	Tier2	Uncertain significance	4.9	—
USP47	c.1519-181G>C	chr11:11929863	rs140531317	Heterozygous	Tier2	Not reviewed	13.4	—
VAMP1	c.130-6C>G	chr12:6466006	rs781012325	Heterozygous	Tier2	Uncertain significance	4.9	—
VAPB	c.*5250T>C	chr20:58449485	rs6015274	Heterozygous	Tier2	Benign	9.5	—
VPS13B	p.Arg1740fs	chr8:99639998	chr8:99639998	Homozygous	Tier2	Not reviewed	—	—
VPS13B	p.Thr1869Met	chr8:99642196	rs117148013	Heterozygous	Tier2	Conflicting classifications of pathogenicity	5.6	0.041
VPS13B	p.Asn2139Ser	chr8:99699894	rs142248228	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.0	0.061
VPS13C	p.Ser2904Leu	chr15:61911844	rs115869241	Heterozygous	Tier2	Benign/Likely benign	25.4	0.233
VPS13C	c.6975+93A>G	chr15:61922304	rs77733354	Heterozygous	Tier2	Benign	0.6	—
VPS13D	p.Ser2684=	chr1:12327709	rs28687897	Heterozygous	Tier2	Benign	6.0	—
VPS35L	p.Thr518=	chr16:19629820	rs144908771	Heterozygous	**Tier1**	Likely benign	22.5	—
VWA3B	p.Glu372Ter	chr2:98162976	rs62154921	Heterozygous	**Tier1**	Benign	46.0	—
VWF	c.658-1399A>G	chr12:6076950	rs7135976	Heterozygous	Tier2	Not reviewed	4.7	—
WASHC4	p.Tyr38His	chr12:105111175	rs35294333	Heterozygous	Tier3	Benign	22.8	0.145
WDHD1	p.Val210Ala	chr14:55002157	rs374497720	Heterozygous	**Tier1**	Not reviewed	25.8	0.271
WDR62	p.Val85Met	chr19:36058855	rs146274964	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.8	0.154

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XDH	p.Ile646Val	chr2:31370399	rs17323225	Heterozygous	Tier3	Benign/Likely benign	0.0	0.008
ZAN	p.His1922Tyr	chr7:100773850	rs314299	Heterozygous	Tier2	Benign	18.2	—
ZAP70	c.1737-3C>T	chr2:97739372	rs56249179	Heterozygous	Tier2	Conflicting classifications of pathogenicity	14.9	—
ZBTB11	c.1801-886G>T	chr3:101660914	rs116667442	Heterozygous	Tier2	Not reviewed	1.5	—
ZBTB20	c.-255+46175_-255+46176del	chr3:114454210	chr3:114454210	Homozygous	Tier2	Not reviewed	—	—
ZFHX3	p.Gln2759His	chr16:72794405	rs2229288	Heterozygous	Tier2	Benign	19.3	0.393
ZFHX3	c.-1705_-1696dup	chr16:73891745	chr16:73891745	Heterozygous	Tier2	Not reviewed	—	—
ZNF808	p.Arg494His	chr19:52554397	rs149957912	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.1	0.007
ZNRF3	p.His637Arg	chr22:29050091	rs62641746	Heterozygous	Tier2	Not reviewed	0.6	0.024
ZRSR2	p.Ser447_Arg448dup	chrX:15823108	chrX:15823108	Heterozygous	Tier3	Likely benign	—	—
ZSWIM6	p.Ala147_Gly153del	chr5:61332694	rs1186750536	Heterozygous	Tier2	Uncertain significance	—	—
ZYX	c.1024-1941C>G	chr7:143386278	rs531543959	Heterozygous	Tier2	Not reviewed	3.8	—